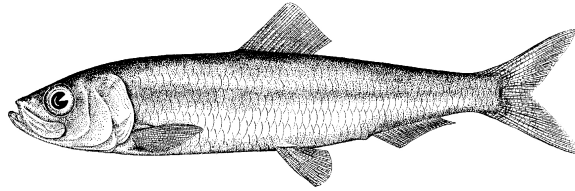


Pacific Herring preliminary data summary for Central Coast 2025

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Pacific Herring (*Clupea pallasii*). Image credit: [Fisheries and Oceans Canada](#).

Disclaimer This report contains preliminary data collected for Pacific Herring in 2025 in the Central Coast major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2025. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

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stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2025 in the Central Coast major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). For the Central Coast major SAR, we use another level of spatial aggregation which we refer to as a ‘Group’. Note that we refer to ‘year’ instead of ‘herring season’ in this report; therefore 2025 refers to the 2024/2025 Pacific Herring season.

2 Data collection programs

The data collection program in the Central Coast reflected a number of collaborations between DFO, the Heiltsuk Nation, and the Herring Industry. Biological samples were collected by the seine test charter vessel *Ocean Venture* for 22 days from March 16th to April 7th. The primary purpose of the test charter vessel was to collect biological samples from main bodies of herring in Statistical Areas 06, 07, and 08, identified from soundings. Heiltsuk and Kitasoo Nations collected nearshore cast net herring samples again this season. Herring spawn locations were primarily identified with fixed-wing overflights. Eight spawn flights were conducted this season between March 21st and April 11th. Additional spawn reconnaissance for Area 08 was conducted by the Wuikinuxv Fisheries Program.

Two dive charter vessels operated in the CC:

- The *Pachena No. 1* surveyed 16 days from April 5th to April 21st, and
- The *Ocean Cloud* surveyed 12 days from April 5th to April 16th.

2.1 Biological samples in Statistical Area 08

In the Central Coast major SAR, Pacific Herring in Statistical Area (SA) 08 tend to be smaller than fish in other areas. In addition, fewer biological samples are typically collected from SA 08 compared to the other areas. For example, between 1994 and 2013, on average $\sim 7\%$ of biological samples came from SA 08. In 2014 and 2015, additional resources were available to collect biological samples in SA 08, which consequently received more sampling effort than previous years, compared to the other areas. In order to avoid using non-representative biological data in the Central Coast stock assessment model, biological samples from SA 08 in 2014 and 2015 are weighted by the average historic proportion of samples in that area (i.e., $\sim 7\%$). Biological data presented in this report (e.g., proportion-at-age, weight-at-age) reflect these weights, and are considered to be representative of the Central Coast major SAR.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2025 in the Central Coast major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated with SOK harvest. See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Incidental mortality from fisheries and aquaculture activities is shown in Figure 4.

In 2025, 31 Pacific Herring biological samples were collected and processed for the Central Coast major SAR (Table 5, Table 6). The nearshore sampling program is a multi-year pilot study using cast nets (Figure 10, Figure 11, and Figure 12). Differences between biological data collected from two sampling protocols regarding the number-, proportion-, weight-, and length-at-age for Pacific Herring in 2025 in the Central Coast major SAR are shown in Table 7, Table 8, Table 9, and Table 10, respectively. Summaries of data collected by the nearshore sampling program are shown in Table 11, Table 12, Table 13, and Table 14. We compare length-, weight, and proportion-at-age as well as length-weight relationships for nearshore and seine test samples in Figure 13, Figure 14, Figure 15, and Figure 16, respectively. Note that only biological data from the seine samples are used for the purposes of stock assessment. The locations in which the biological samples were collected are presented in Figure 5. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 6, Table 15, and Figure 7, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 8 & Figure 9, respectively). Some Statistical Areas tend to have larger fish at a given age (Figure 17, Table 16). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where ‘representative’ indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

3.1 Nearshore genetics project

DFO initiated a nearshore genetics project in 2022 to better understand the spatial and temporal structure of Pacific Herring stocks in British Columbia. We do this by

collecting genetic and biological data from actively spawning Pacific Herring (*Clupea pallasii*). Then we collect biological data (e.g., length, weight, age, sex) and genetic data from the fish. Finally, we analyse the biological and genetic data to better understand Pacific Herring. Genetic samples were collected in the Central Coast major SAR in 2023 and 2024 (Figure 18). Results from the nearshore genetics project will be presented in a separate document when they are available.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 51 individual locations in 2025 in the Central Coast major SAR (Table 17, and Figure 19). A summary of spawn from the last decade (2015 to 2024) is shown in Figure 20. Figure 21 shows spawn start date by decade and Statistical Area. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 22, Figure 23, Figure 24, Figure 25, Table 18, and Figure 26). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. In addition, spawn surveys estimate spawn depth by Statistical Area, and Section (Figure 27). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 25b, Figure 28). For example, in 2025, Section 074 contributed the most to the spawn index (43%). As with Sections, some Statistical Areas contribute more than others to the total spawn index (Figure 25c, Figure 29). An animation shows the spawn index by spawn survey location from 1951 to 2025 (Figure 30).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make minor edits for clarity and brevity, but we do not change the intent or substance of responses.

5.1 Wuikinuxv

Lack of FSC herring this year and less spawn observed this year compared to the last three years.

5.2 Kitasoo Xai'xais

Active spawn was first observed by KX Stewardship Authority staff in the territory at Marvin Island in Kitasu Bay on March 24th, following the ponding of live fish for the commercial SOK harvest. Natural spawn then built gradually along shorelines of Kitasu Bay into Osment Inlet and Parson's Anchorage, and then along the Wilby Point shoreline for the following week. Active spawn was also observed again this season in Higgins Passage on March 31st and developed into a relatively extensive coverage of north and south shorelines. Active spawn was also observed this season at several locations on West Aristazabal Island, and field staff were able to collect genetic samples from Clifford Bay on April 2nd which is a rarely sampled location. Several smaller spawns were observed later in the season outside of these main spawning areas in the territory, including several locations in Meyers Passage, as well as a small annual deepwater spawn in Kynoch Inlet. The Meyers Passage spawns are gradually building following concerted effort by community harvesters to transplant spawn on hemlock branches from larger adjacent harvesting sites.

Although sounding surveys revealed that mature herring were fairly abundant in the area leading up to the spawning season, community harvesters reported a relatively contracted spawn extent, duration, and density. Harvesters attributed this to increased humpback whale activity in Kitasu Bay and Higgins Passage, where up to three whales at a given time were foraging in each of these locations. KX Stewardship Authority staff continued to closely monitor the piscivorous seabird community over the herring spawn season as tens of thousands of ducks, gulls, and alcids aggregated to forage on eggs and spawning fish. Seabird abundance and distribution remained consistent with observations over the past few years, with single-day peak counts of upwards of 40,000 surf scoters observed in Kitasu Bay and Higgins Passage. The peak surf scoter count occurred after peak herring emergence again this season, although thousands of gulls, scoters, and other ducks, and hundreds of murres and loons were observed to be actively foraging on incubating spawn across most of the region.

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- Generally good weather resulted in minimal interference with the assessment program.
- First spawn reported on March 26th in Area 6 (Kitasu Bay).
- Sea surface temperatures through March and the start of April were unusually cold, averaging 5°C.
- Early indications of spawning activity were not favourable. Spawning conditions improved significantly as the season progressed.

- In Area 7, FSC product was generally above average. Notable spawn layers observed in lower Spiller Channel.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas, Sections, and Groups for Pacific Herring in the Central Coast major stock assessment region (SAR). Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

Region	Statistical Area	Section	Group
Central Coast	06	067	06&07
Central Coast	07	070	06&07
Central Coast	07	071	06&07
Central Coast	07	072	06&07
Central Coast	07	073	06&07
Central Coast	07	074	06&07
Central Coast	07	075	06&07
Central Coast	07	076	06&07
Central Coast	07	077	06&07
Central Coast	07	078	06&07
Central Coast	07	079	06&07
Central Coast	08	085	08
Central Coast	08	086	08

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2025 in the Central Coast major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	0
RoeSN	0
RoeGN	0

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2015 to 2025 in the Central Coast major stock assessment region (SAR). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Harvest does not include the 2,411 lbs of spawn on egreia in 2018. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2015	169,470	251
2016	351,953	521
2017	392,747	582
2018	289,358	429
2019	356,042	528
2020	44,857	66
2021	294,269	436
2022	0	0
2023	42,022	62
2024	30,712	46
2025	44,401	66

Table 5. Number of Pacific Herring biological samples processed from 2015 to 2025 in the Central Coast major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

Year	Number of samples			
	Commercial	Test	Nearshore	Total
2015	6	14	0	20
2016	5	15	0	20
2017	4	13	27	44
2018	4	12	18	34
2019	2	10	28	40
2020	0	12	5	17
2021	2	13	14	29
2022	0	14	15	29
2023	2	11	5	18
2024	0	12	6	18
2025	0	19	12	31

Table 6. Number and type of Pacific Herring biological samples processed in 2025 in the Central Coast major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Test	Other	Nearshore	12
Test	Seine	Test fishery	19

Table 7. Observed number-at-age of Pacific Herring by sample type in 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Number-at-age								
	2	3	4	5	6	7	8	9	10
Nearshore	27	69	15	27	3	4	4	NA	NA
Seine test	89	565	313	510	84	90	25	19	5
Total	116	634	328	537	87	94	29	19	5

Table 8. Observed proportion-at-age of Pacific Herring by sample type in 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
Nearshore	0.173	0.442	0.096	0.173	0.019	0.026	0.026	NA	NA
Seine test	0.052	0.332	0.184	0.300	0.049	0.053	0.015	0.011	0.003
Total	0.062	0.342	0.177	0.289	0.047	0.051	0.016	0.010	0.003

Table 9. Observed mean weight-at-age in grams (g) of Pacific Herring by sample type in 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Mean weight-at-age (g)								
	2	3	4	5	6	7	8	9	10
Nearshore	49	56	66	80	109	78	105	NA	NA
Seine test	54	72	85	100	103	108	124	131	132
Total	53	70	84	99	103	107	121	131	132

Table 10. Observed mean length-at-age in millimetres (mm) of Pacific Herring by sample type in 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Mean length-at-age (mm)								
	2	3	4	5	6	7	8	9	10
Nearshore	149	157	166	177	195	179	193	NA	NA
Seine test	148	164	174	182	184	187	193	198	200
Total	148	164	174	182	184	187	193	198	200

Table 11. Observed number-at-age of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Number-at-age								
	2	3	4	5	6	7	8	9	10
2017	97	279	208	1111	174	454	85	74	18
2018	24	328	252	231	521	135	87	33	10
2019	15	1543	605	133	129	100	26	7	3
2020	31	10	301	66	29	23	4	4	1
2021	9	226	51	833	134	28	20	8	1
2022	51	177	308	43	527	68	19	6	3
2023	12	238	63	48	15	81	15	1	1
2024	4	29	146	14	17	11	17	2	0
2025	27	69	15	27	3	4	4	0	0

Table 12. Observed proportion-at-age of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
2017	0.039	0.112	0.083	0.444	0.070	0.182	0.034	0.030	0.007
2018	0.015	0.202	0.155	0.143	0.321	0.083	0.054	0.020	0.006
2019	0.006	0.602	0.236	0.052	0.050	0.039	0.010	0.003	0.001
2020	0.066	0.021	0.642	0.141	0.062	0.049	0.009	0.009	0.002
2021	0.007	0.173	0.039	0.636	0.102	0.021	0.015	0.006	0.001
2022	0.042	0.147	0.256	0.036	0.438	0.057	0.016	0.005	0.002
2023	0.025	0.502	0.133	0.101	0.032	0.171	0.032	0.002	0.002
2024	0.017	0.121	0.608	0.058	0.071	0.046	0.071	0.008	0.000
2025	0.173	0.442	0.096	0.173	0.019	0.026	0.026	0.000	0.000

Table 13. Observed mean weight-at-age in grams (g) of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Mean weight-at-age (g)								
	2	3	4	5	6	7	8	9	10
2017	49	69	83	93	98	115	116	120	128
2018	48	68	78	94	96	104	107	107	109
2019	52	61	67	84	97	103	102	119	114
2020	34	59	73	78	89	88	102	89	93
2021	38	58	74	82	91	94	103	120	104
2022	46	60	72	92	94	103	108	121	108
2023	43	64	74	85	99	104	109	111	107
2024	36	64	79	83	100	100	98	106	NA
2025	49	56	66	80	109	78	105	NA	NA

Table 14. Observed mean length-at-age in millimetres (mm) of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Mean length-at-age (mm)								
	2	3	4	5	6	7	8	9	10
2017	158	175	187	193	198	207	208	209	213
2018	154	172	178	191	193	199	201	204	200
2019	153	162	167	180	190	195	196	200	206
2020	136	162	173	178	186	184	193	187	194
2021	141	161	175	180	185	192	196	200	195
2022	150	164	176	190	193	198	201	210	203
2023	145	162	170	178	186	190	191	201	194
2024	142	168	181	185	194	196	199	194	NA
2025	149	157	166	177	195	179	193	NA	NA

Table 15. Observed proportion-at-age for Pacific Herring from 2015 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

Year	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
2015	0.008	0.454	0.156	0.249	0.055	0.064	0.009	0.004	0.001
2016	0.040	0.091	0.549	0.094	0.146	0.028	0.039	0.011	0.003
2017	0.029	0.170	0.132	0.446	0.089	0.092	0.024	0.012	0.007
2018	0.090	0.168	0.174	0.147	0.295	0.066	0.041	0.014	0.005
2019	0.012	0.616	0.178	0.053	0.053	0.059	0.024	0.002	0.002
2020	0.028	0.028	0.739	0.101	0.045	0.029	0.021	0.007	0.002
2021	0.012	0.295	0.031	0.521	0.085	0.033	0.015	0.007	0.001
2022	0.105	0.163	0.376	0.045	0.243	0.047	0.013	0.006	0.002
2023	0.030	0.614	0.118	0.104	0.027	0.089	0.014	0.003	0.000
2024	0.020	0.175	0.549	0.082	0.082	0.032	0.045	0.010	0.005
2025	0.052	0.332	0.184	0.300	0.049	0.053	0.015	0.011	0.003

Table 16. Sample size for Pacific Herring weight-at-age analysis in 2025 in the Central Coast major stock assessment region (SAR) by Group from the most recent decade (2016 to 2025), and the previous decade (2006 to 2015), as displayed in Figure 17. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

Group	Age	Sample size	
		Previous decade	Recent decade
06&07	2	727	483
06&07	3	6,277	2,686
06&07	4	7,729	3,450
06&07	5	3,329	2,464
06&07	6	2,242	1,352
06&07	7	797	628
06&07	8	265	303
06&07	9	97	104
06&07	10	32	38
08	2	265	79
08	3	1,092	579
08	4	1,036	461
08	5	300	203
08	6	179	140
08	7	64	59
08	8	13	26
08	9	5	10
08	10	3	3

Table 17. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2025 in the Central Coast major stock assessment region (SAR). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
06	067	Abrams Is	March 31	871
06	067	Clifford Bay	March 31	159
06	067	Kwakwa Cr	April 01	341
06	067	Larkin Pt	March 31	247
06	067	Marvin Is	March 31	865
06	067	Meyers Nrws	April 08	3
06	067	Meyers Psg	April 08	466
06	067	Normansell Is	March 31	7
06	067	Osment Inlt	March 31	1,188
06	067	Parsons Anch	March 31	824
06	067	Wilby Pt	March 25	1,291
06	067	Wingate Pt	April 01	19
06	067	Woodcock Is	March 31	97
07	072	Branks It	March 31	0
07	072	Don Lgn	March 31	365
07	072	Ivory Is	March 31	53
07	072	Mosquito Bay	March 27	2,447
07	072	Port Blackney	March 31	1
07	072	Powell Anch	March 31	29
07	072	Spiller Chnl	March 31	1,689
07	072	Tankeeah River	April 03	148
07	072	Watch Is	March 31	18
07	074	Agnew Islet	April 03	0
07	074	Cape Swaine	March 31	3,744
07	074	Codfish Pass	April 03	270
07	074	Dundivan Inlt	April 03	707
07	074	Edwards Pt	April 02	157
07	074	Idol Pt	March 27	5,612
07	074	Kimlock Pt	April 03	272
07	074	Lockhart Bay	April 05	11
07	074	McGown Pt	April 04	17
07	074	Rait Nrws	April 03	1,128
07	074	Stryker Is	April 03	1,873
07	074	Wasakesiu Pass	April 02	23
07	075	McNaughton Grp	April 03	45
07	076	Hurricane Is	March 31	52
07	076	Leckie Bay	March 31	32
07	077	E Higgins Pass	March 31	1,100

<i>Table 17 continued</i>				
Statistical Area	Section	Location name	Start date	Spawn index (t)
	07	078 Neekas Cv	March 27	2,258
	07	078 Neekas Inlt	March 27	2,943
	07	078 Spiller Chnl Hd	March 27	33
	08	085 Keith Anch	April 06	50
	08	085 Kwakume Pt	March 27	54
	08	085 Pruth Bay	April 06	124
	08	085 Whittaker Pt	April 06	221
	08	086 Harlequin Basin	March 26	3
	08	086 Hergest Pt	April 02	11
	08	086 Mustang Bay	April 02	36
	08	086 Nalau Psg	April 02	576
	08	086 Stirling Isl	April 02	2
	08	086 Target Bay	April 02	0

Table 18. Summary of Pacific Herring spawn survey data from 2015 to 2025 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2015	167,060	39	1.3	32,145
2016	164,575	40	1.6	32,507
2017	125,525	39	1.5	23,517
2018	98,149	25	1.1	12,263
2019	181,750	37	1.8	46,254
2020	160,625	39	2.0	42,713
2021	128,625	41	1.5	28,674
2022	153,675	42	1.3	22,711
2023	117,250	45	1.4	17,550
2024	137,400	40	1.4	26,802
2025	134,375	41	1.4	32,503

8 Figures

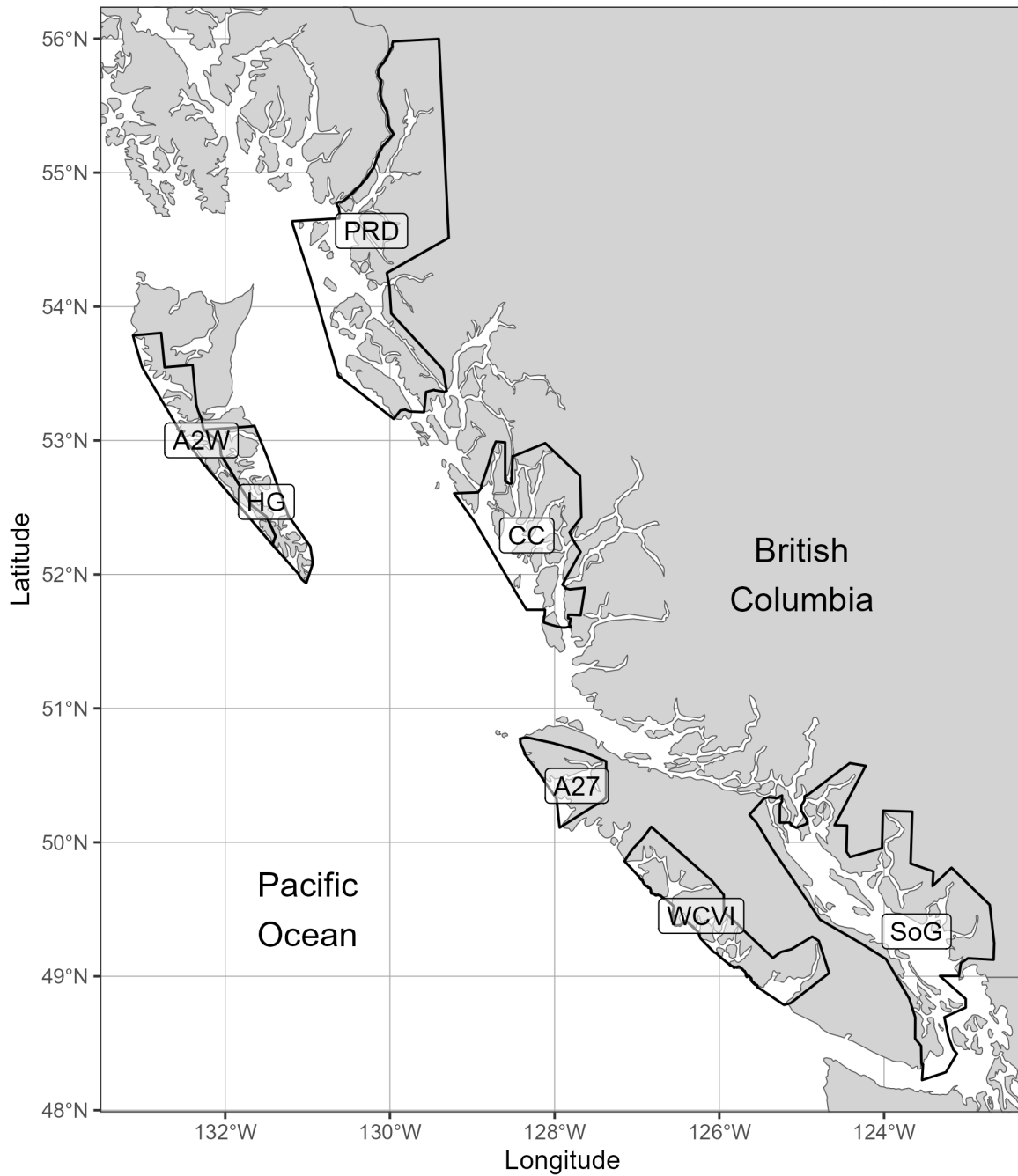


Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

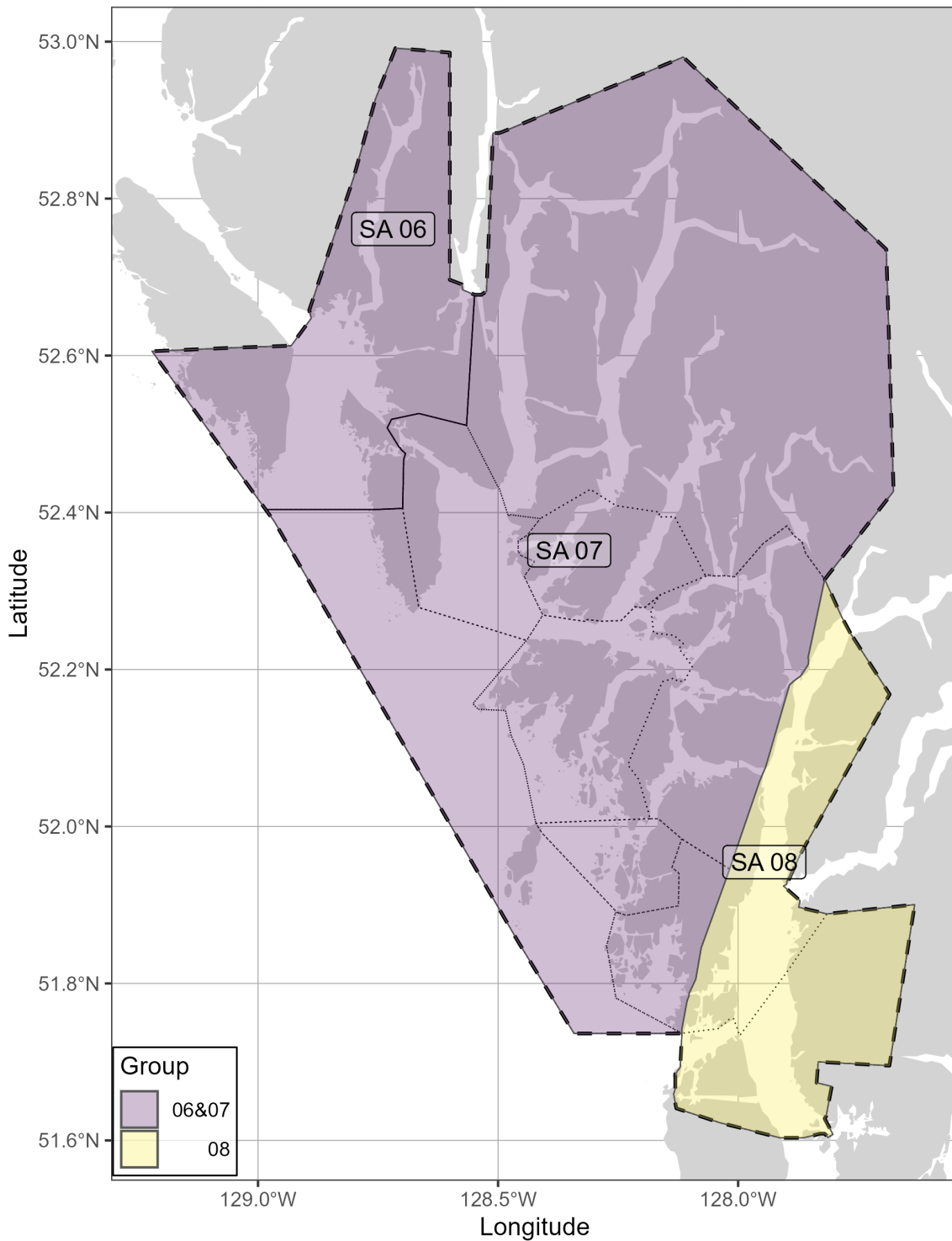


Figure 2. Boundaries for the Central Coast major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km). Legend: '06&07' is Statistical Areas 06 and 07; and '08' is Statistical Area 08.

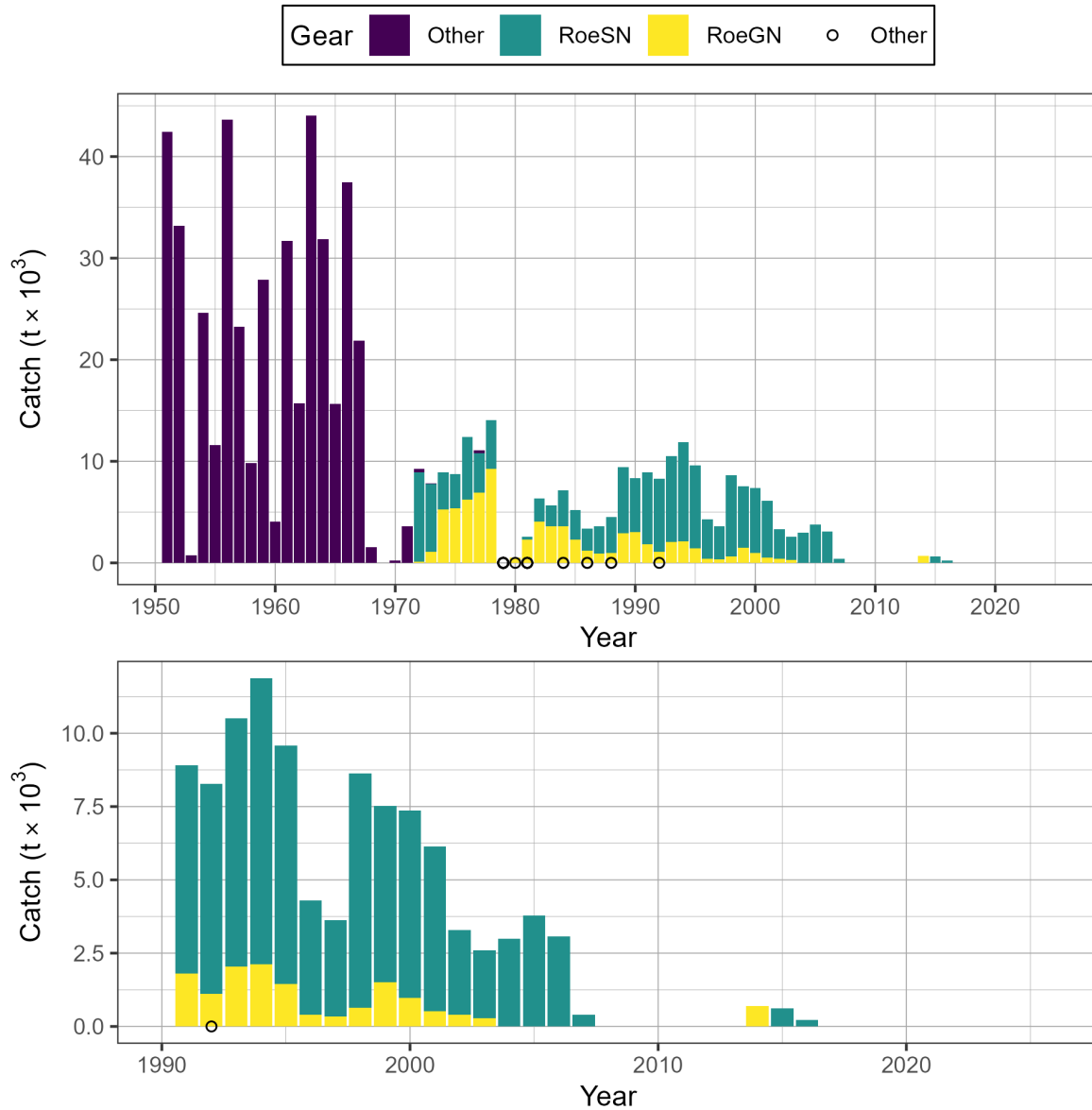


Figure 3. Time series of total landed catch in thousands of metric tonnes ($t \times 10^3$) of Pacific Herring by gear type from 1951 to 2025 in the Central Coast major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1990 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

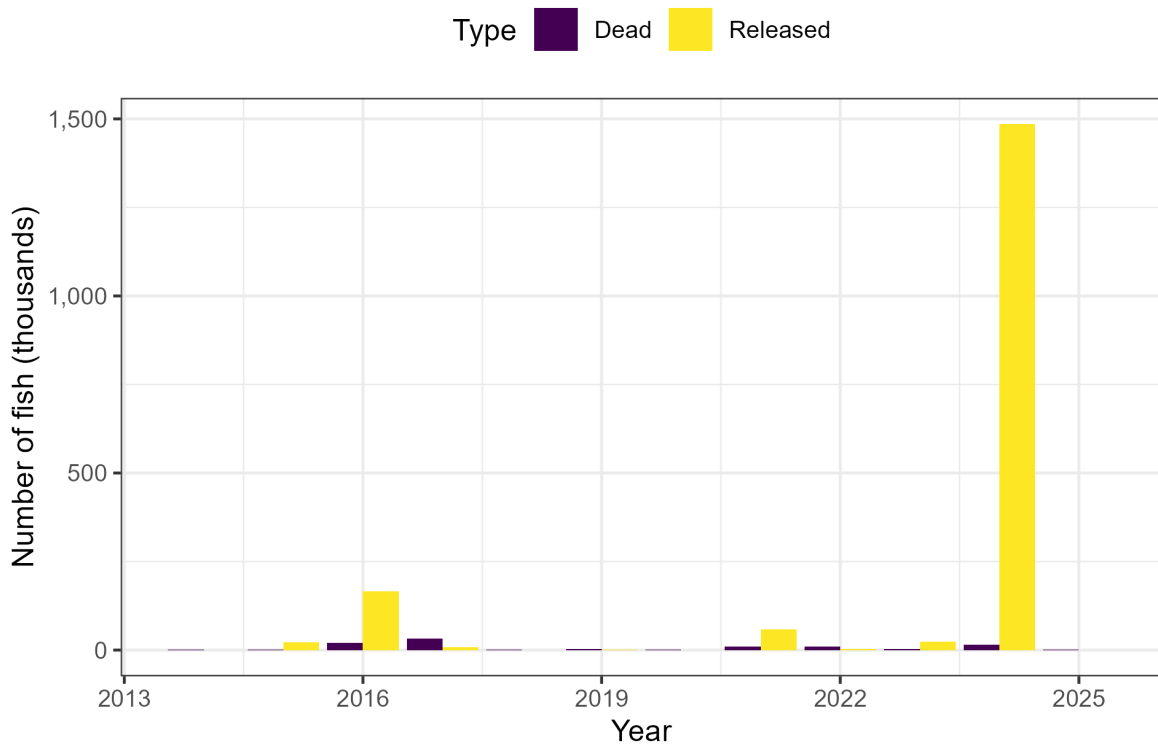


Figure 4. Incidental Pacific Herring mortality in aquaculture activities in thousands of fish from 2014 to 2025 in the Central Coast major stock assessment region (SAR). Note: figure may include data outside SAR boundaries.

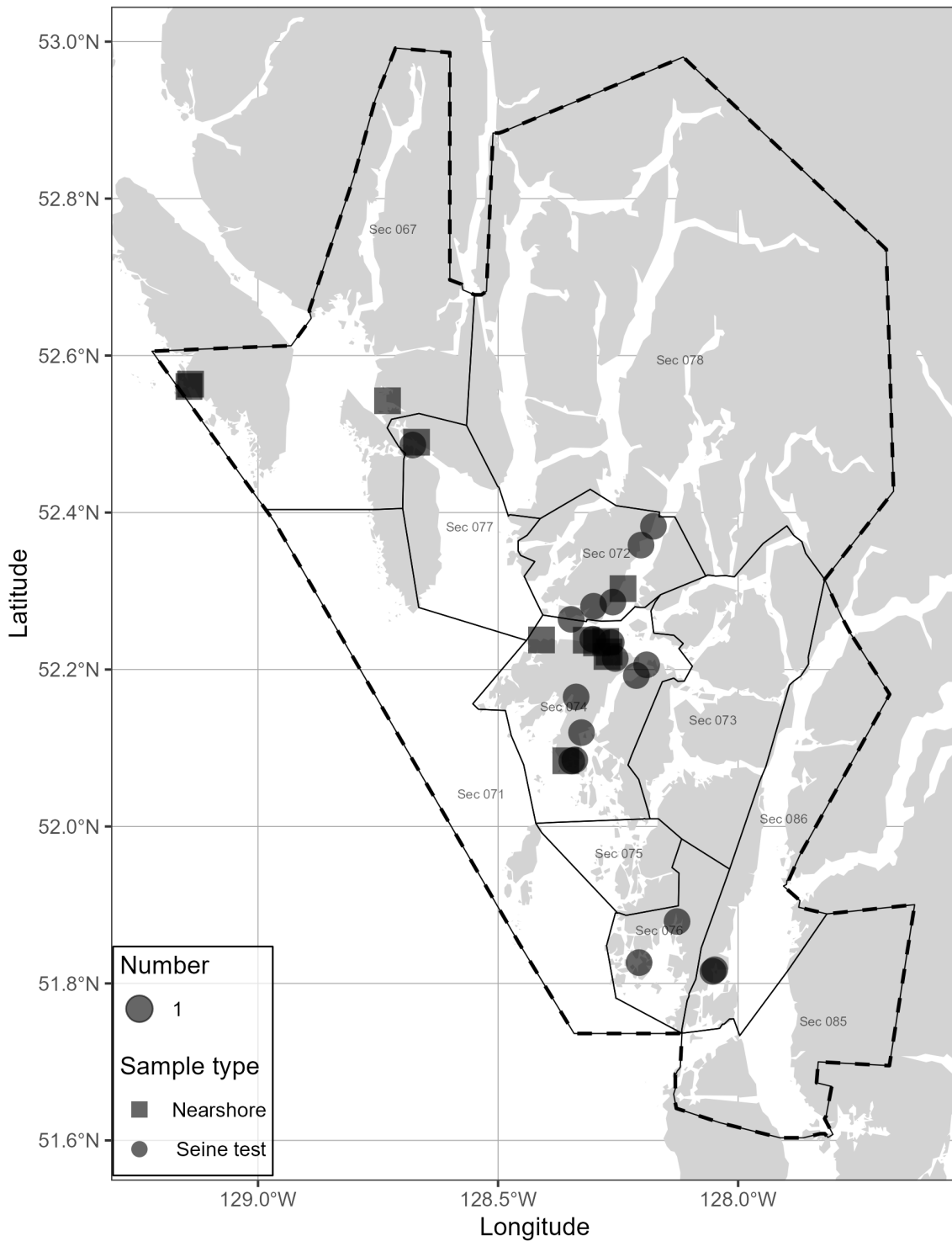


Figure 5. Location and type of Pacific Herring biological samples collected in 2025 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

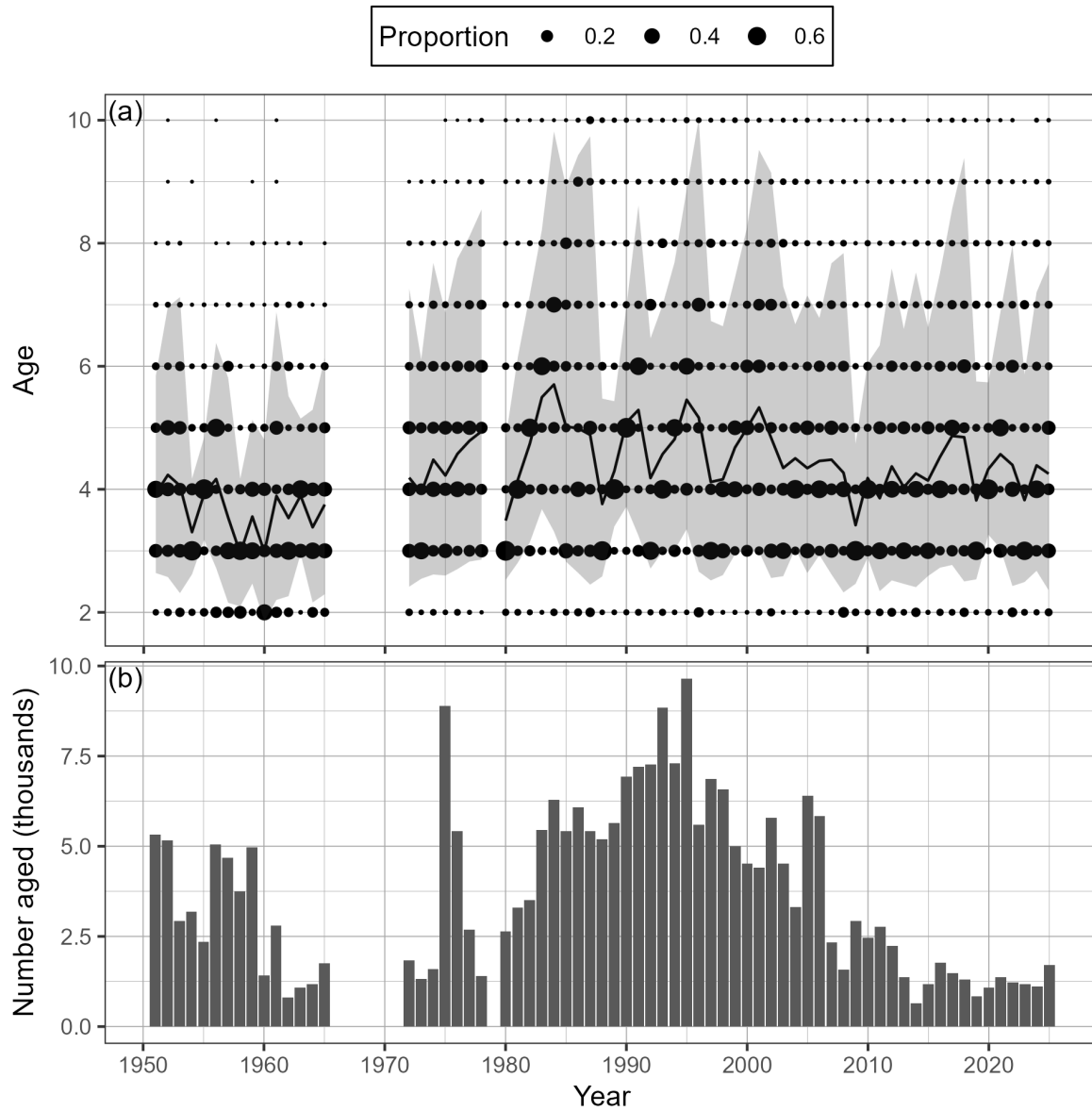


Figure 6. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

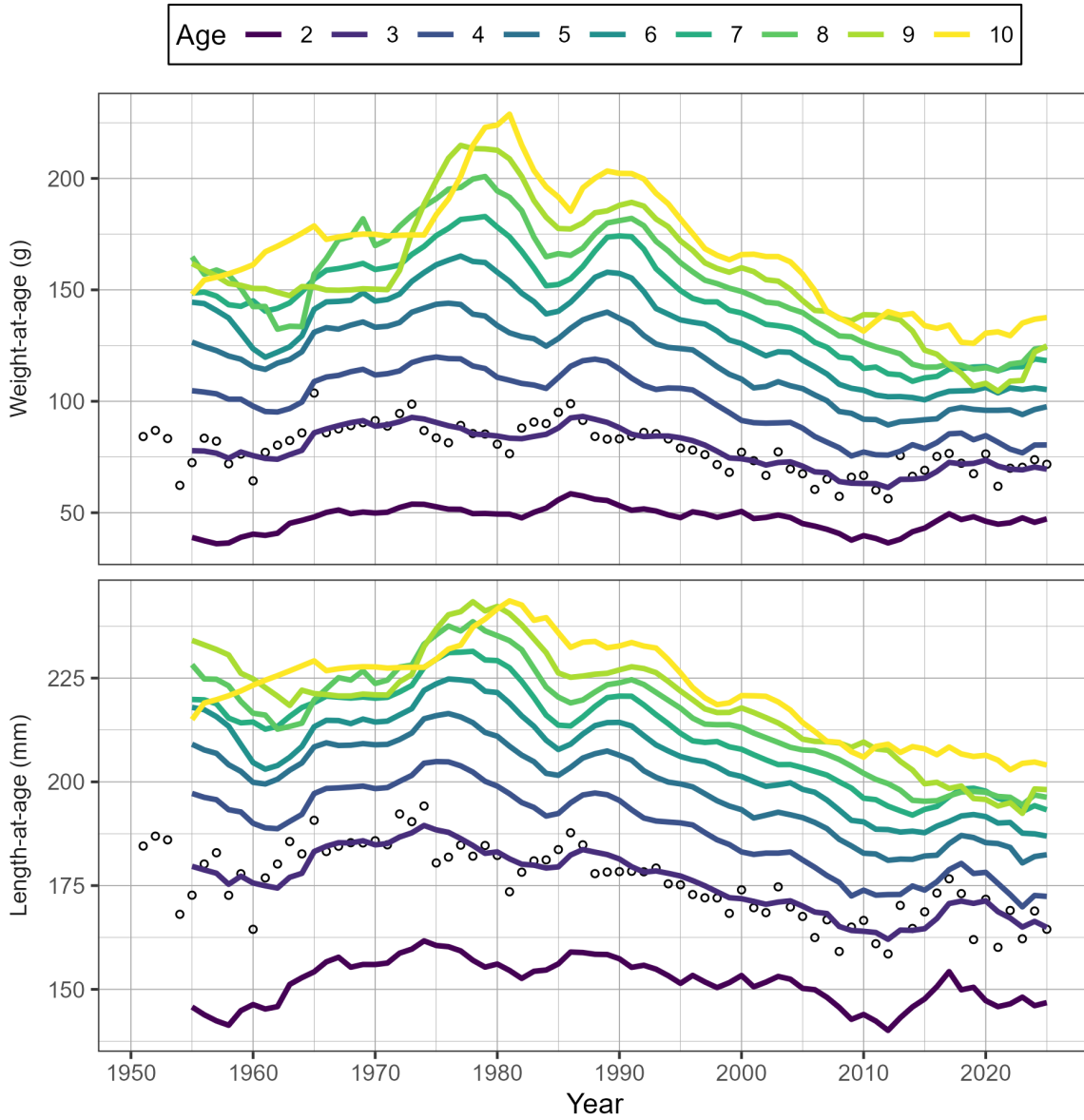


Figure 7. Time series of weight-at-age in grams (g) and length-at-age in millimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

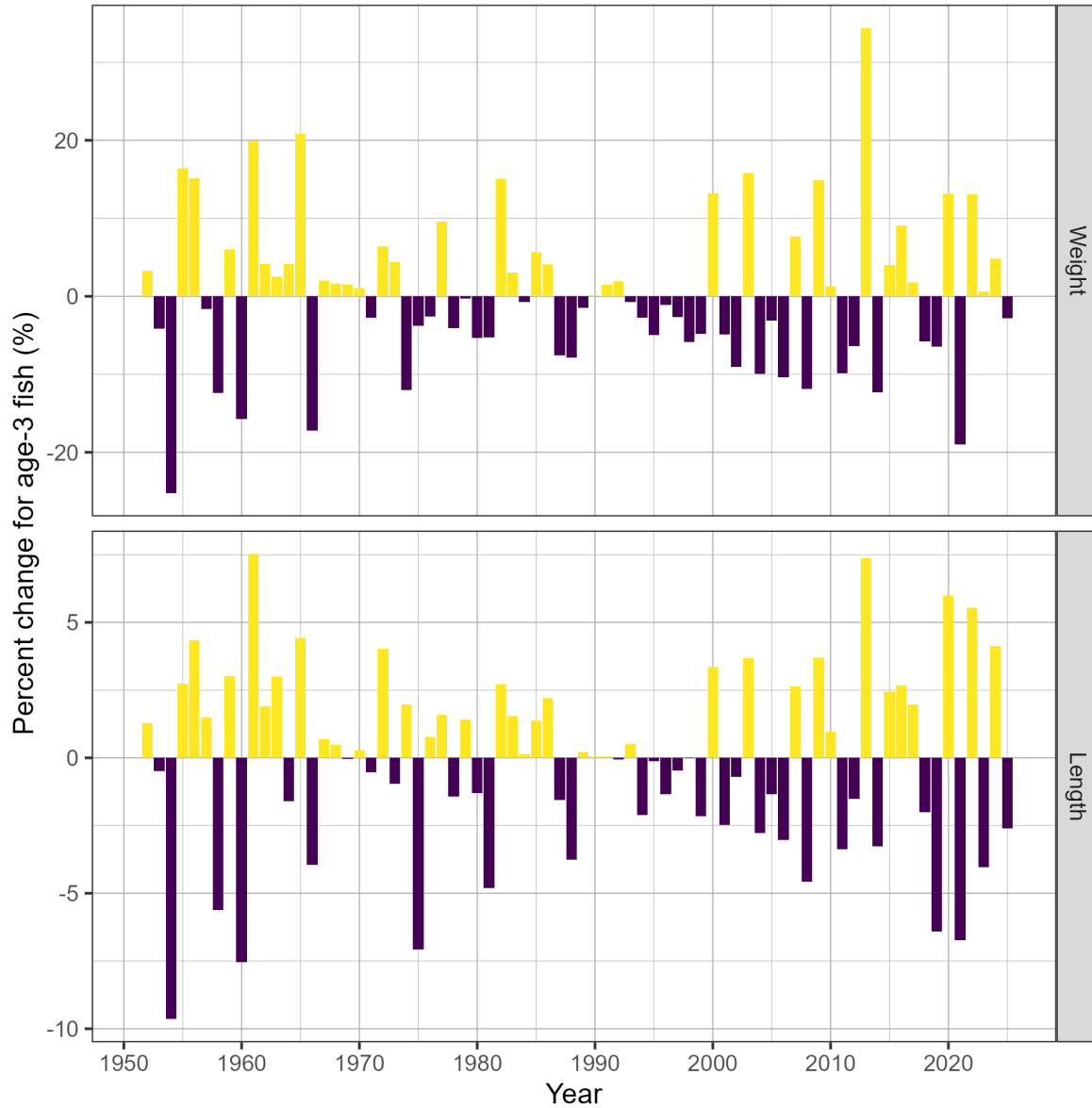


Figure 8. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

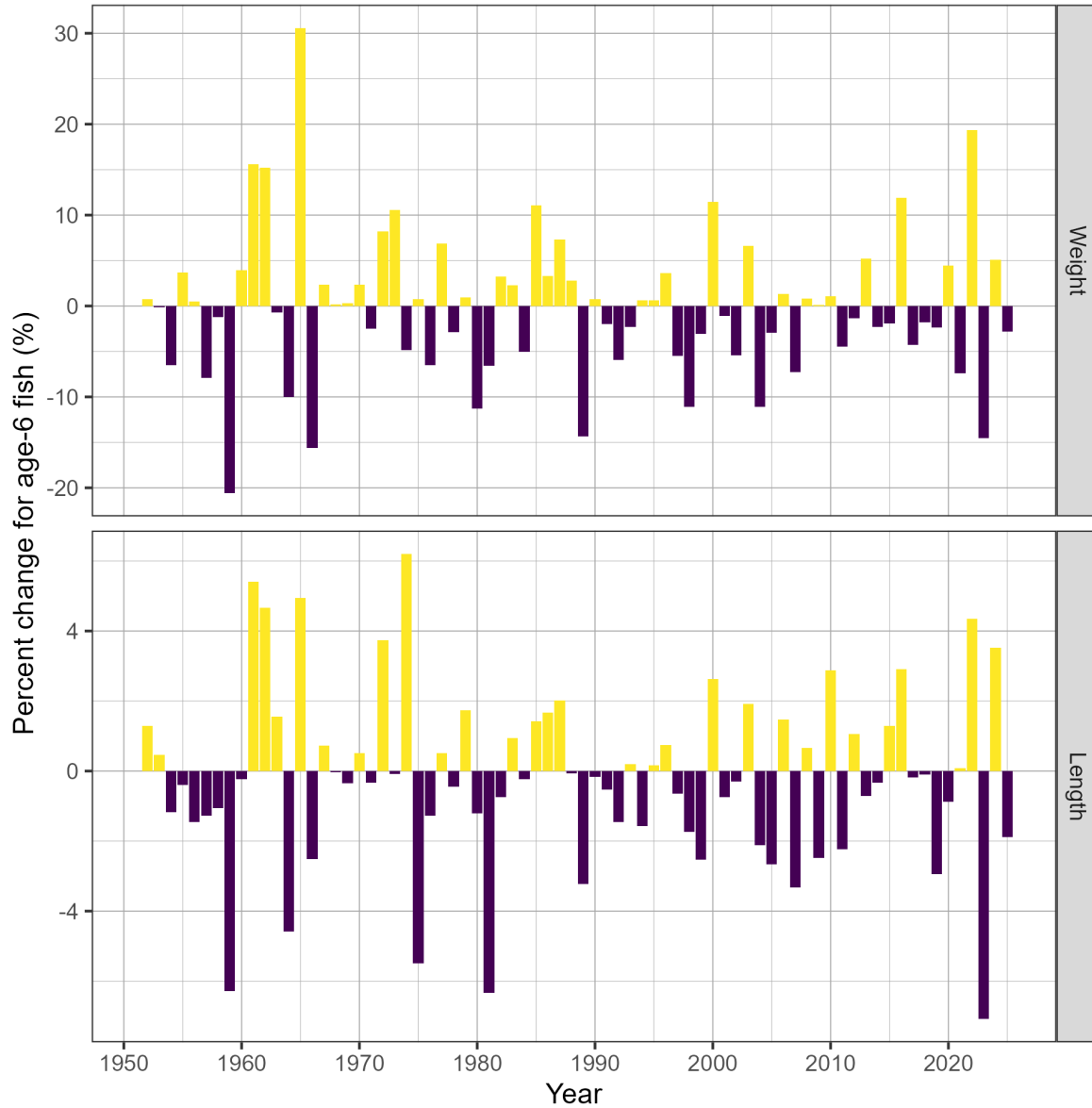


Figure 9. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

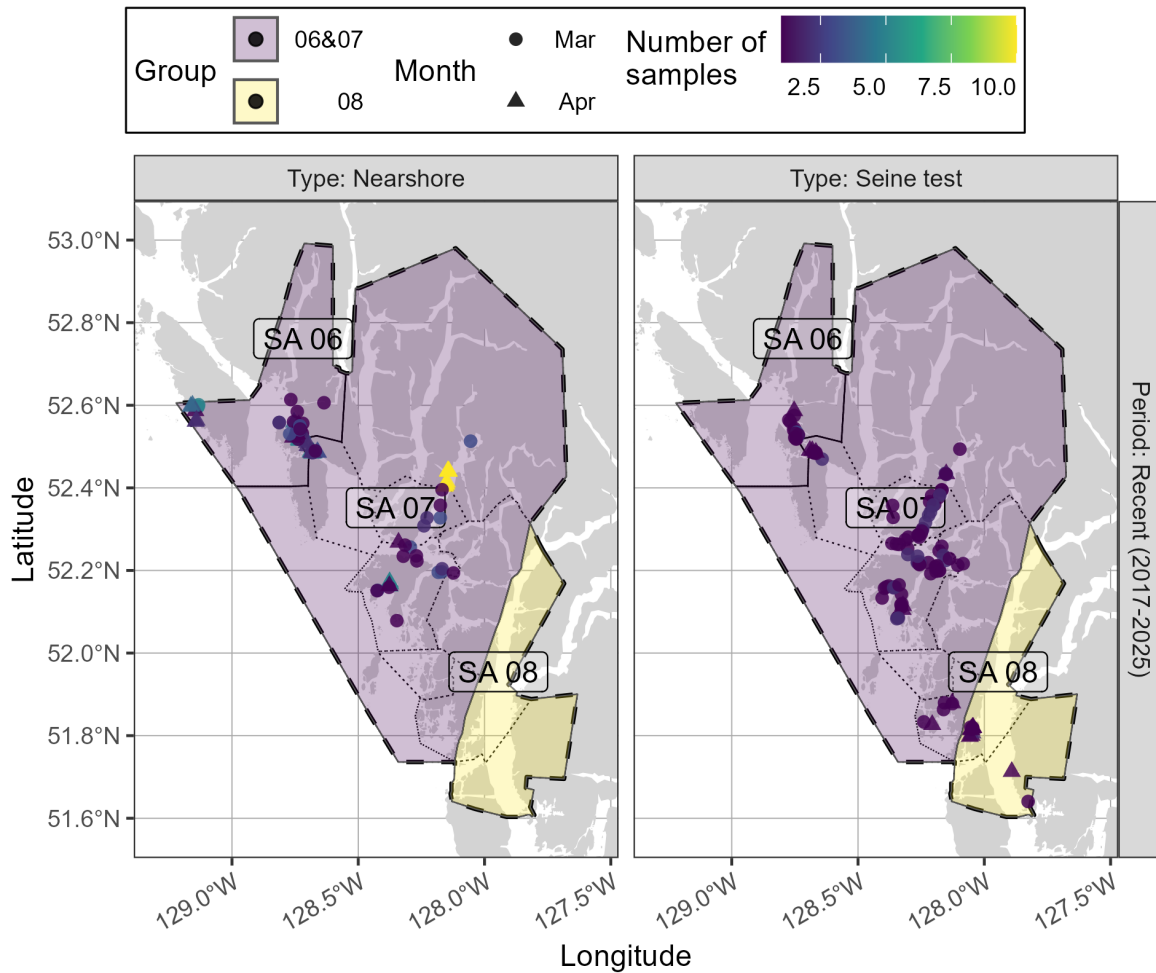


Figure 10. Location of Pacific Herring biological samples in the Central Coast major stock assessment region (SAR) by period, sample type, and month. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

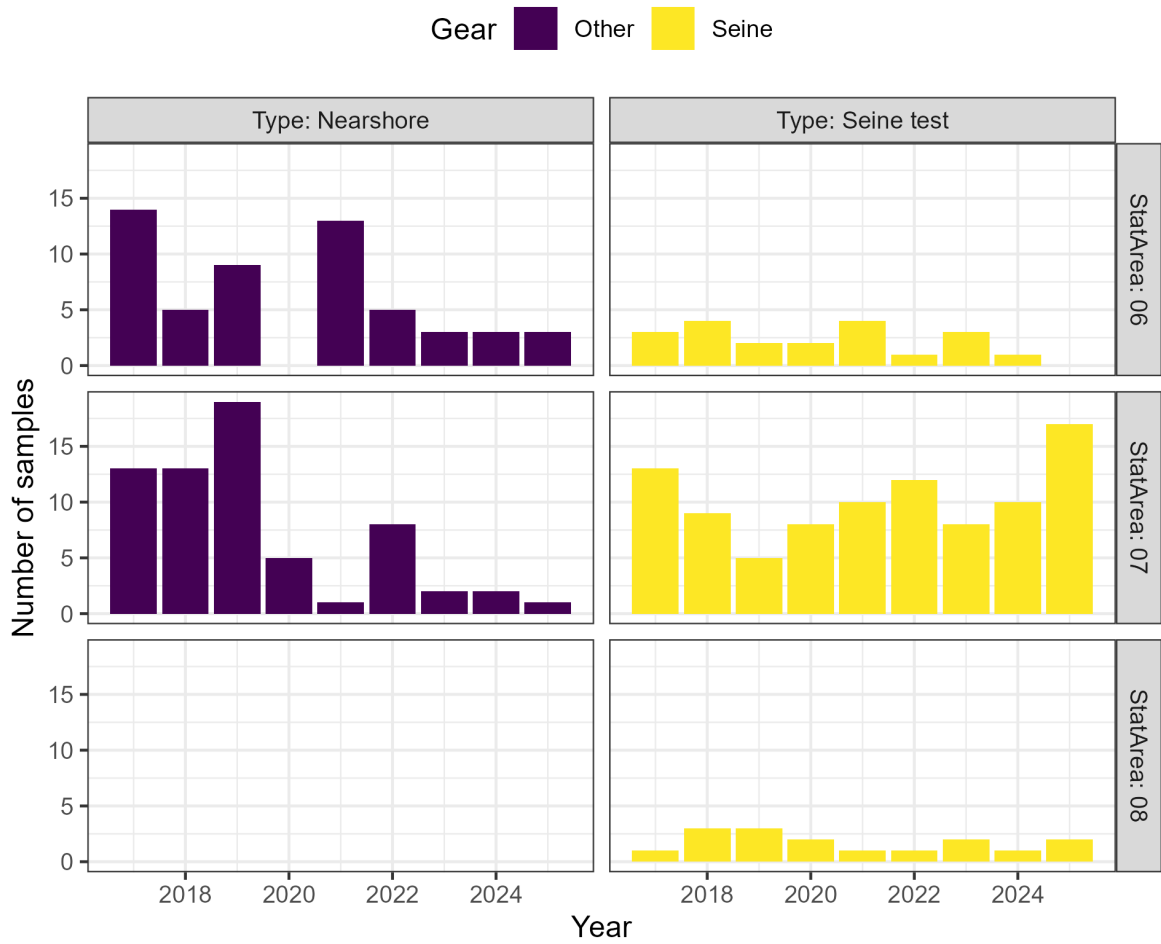


Figure 11. Number of Pacific Herring samples in the Central Coast major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

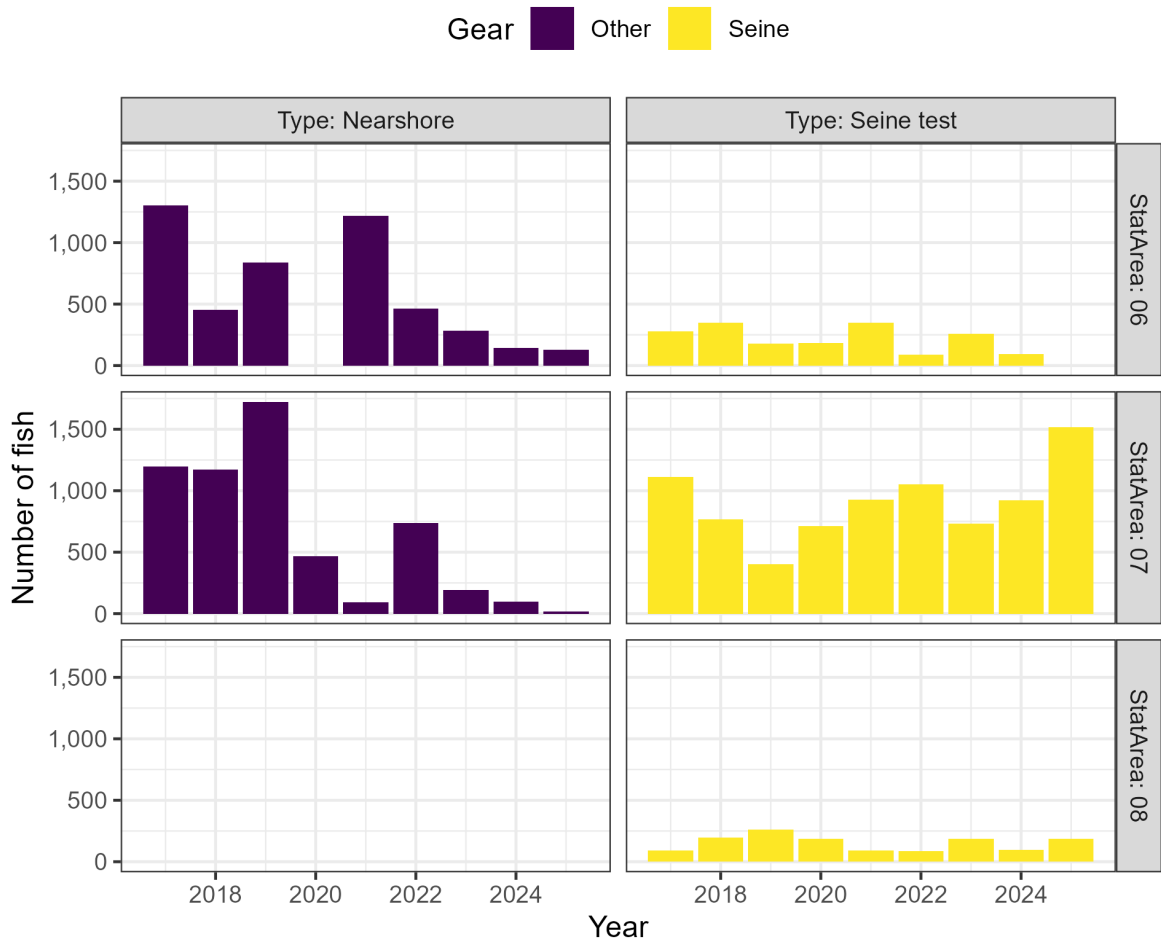


Figure 12. Number of Pacific Herring in biological samples in the Central Coast major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

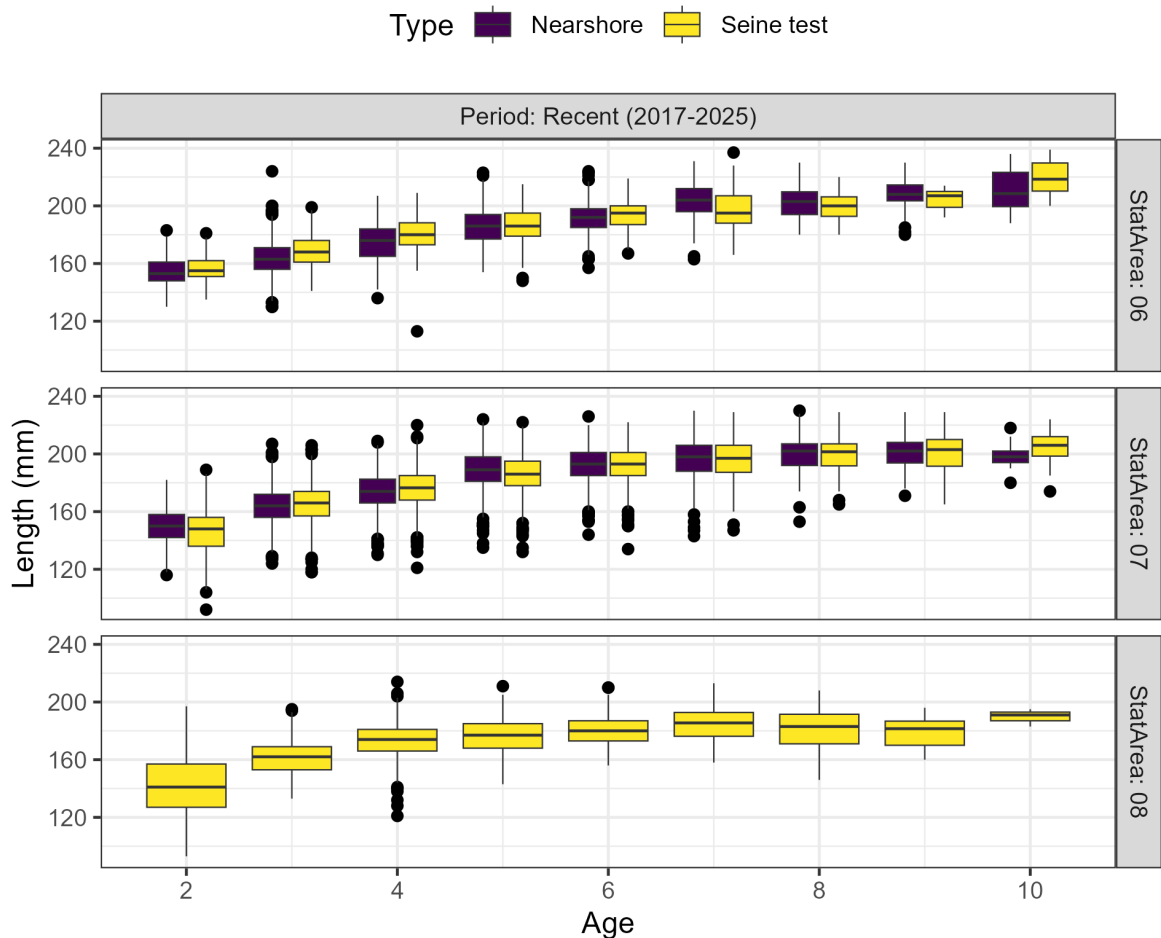


Figure 13. Length-at-age in millimetres (mm) of Pacific Herring in the Central Coast major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

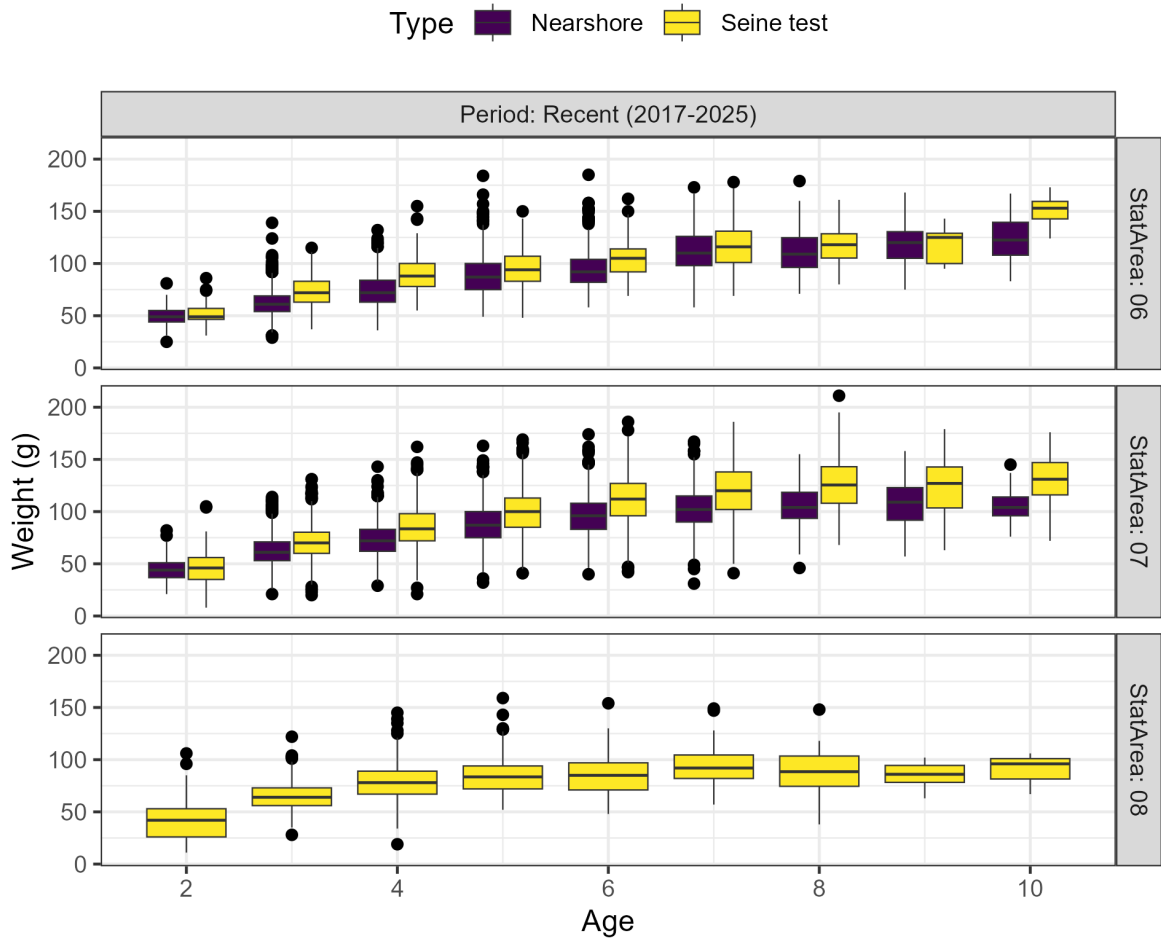


Figure 14. Weight-at-age in grams (g) of Pacific Herring in the Central Coast major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

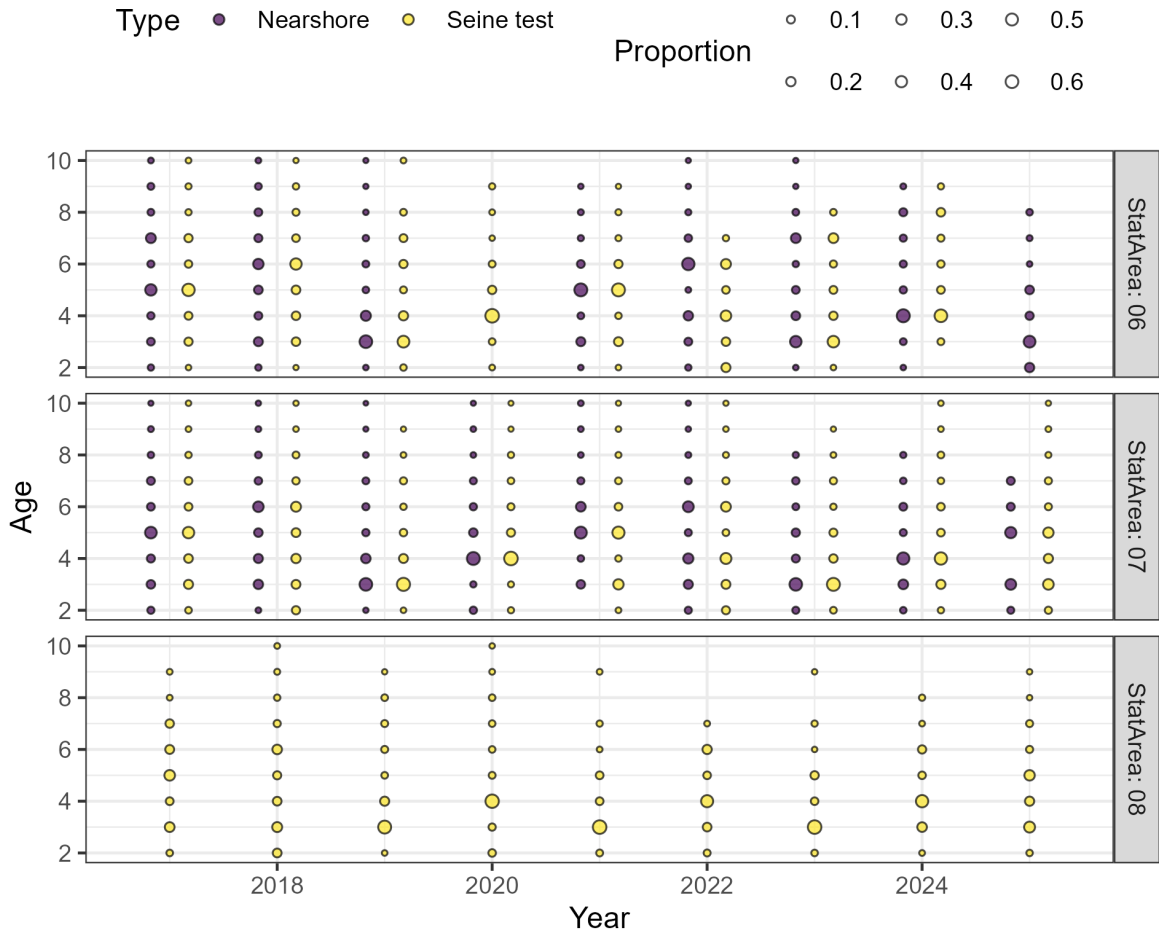


Figure 15. Proportion-at-age of Pacific Herring in the Central Coast major stock assessment region (SAR) by statistical area and sample type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

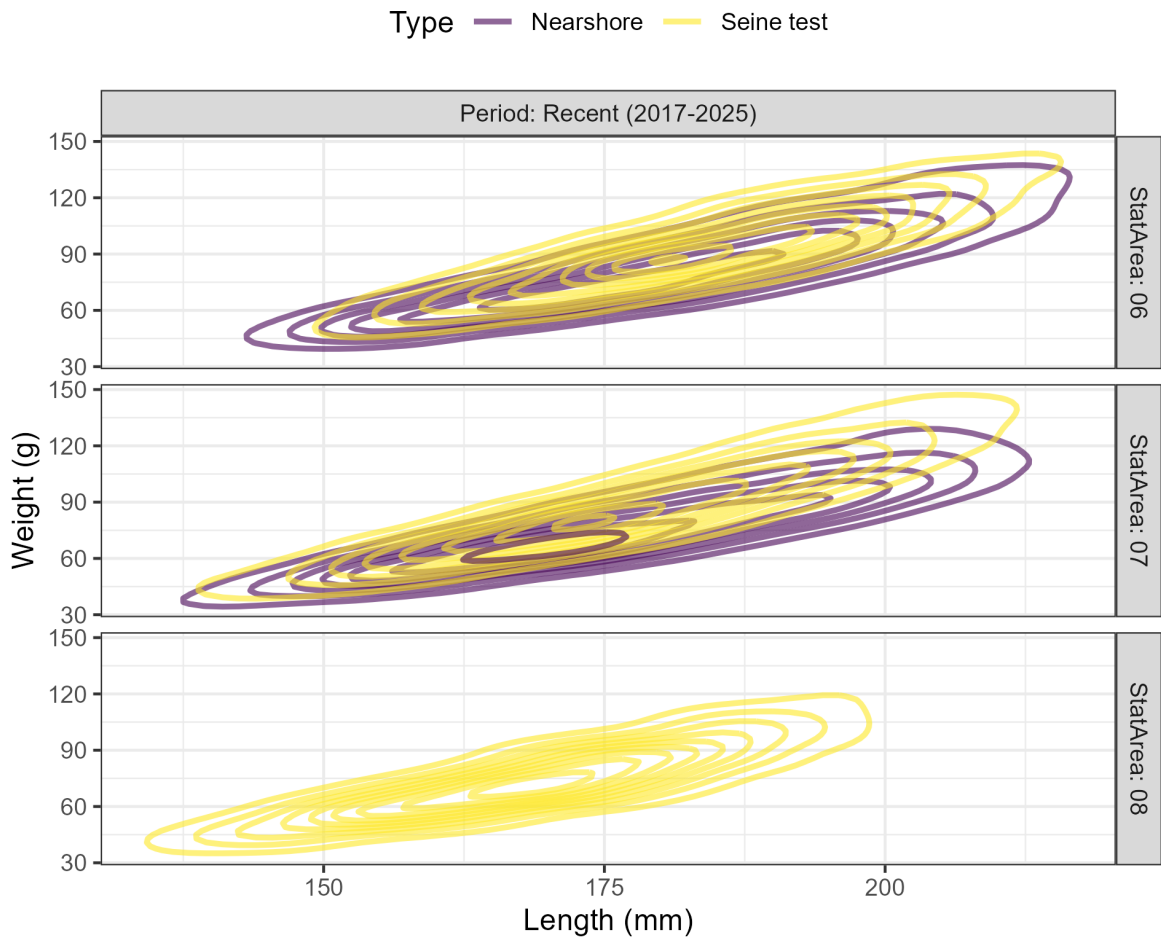


Figure 16. Length-weight relationship for Pacific Herring in the Central Coast major stock assessment region (SAR) by period, statistical area, and sample type. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

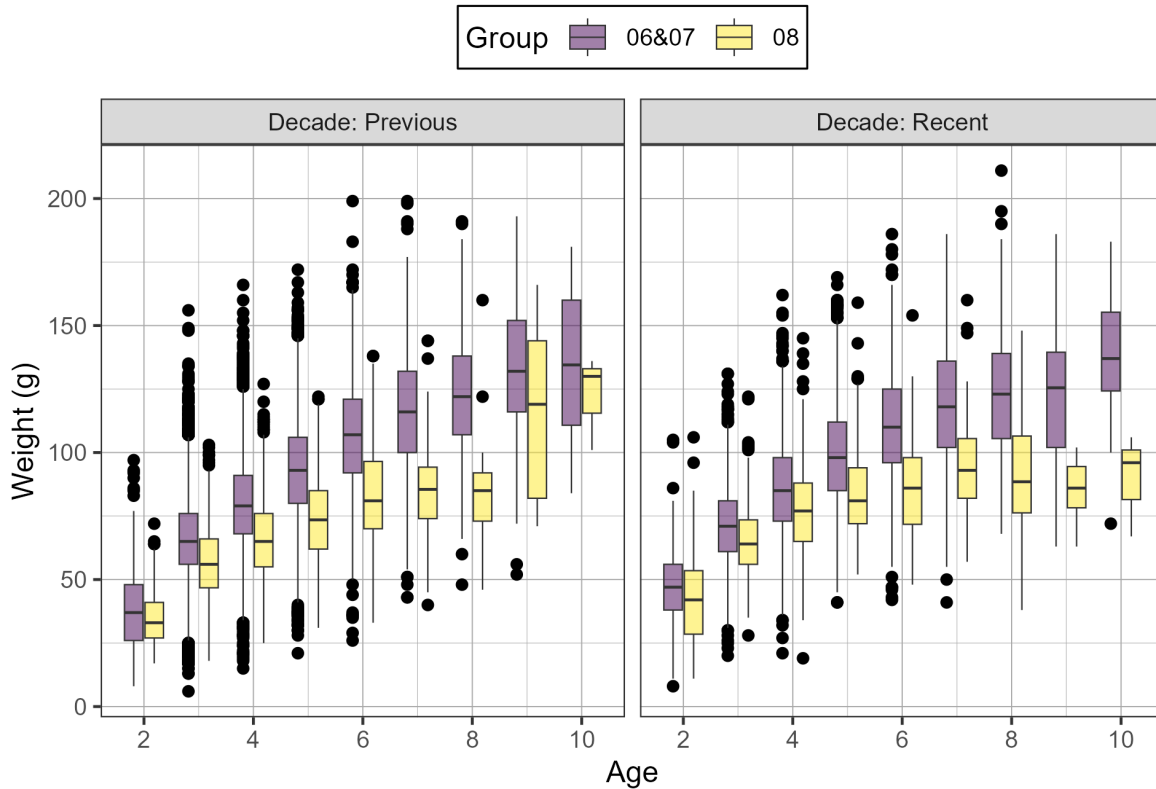


Figure 17. Weight-at-age in grams (g) of Pacific Herring in the Central Coast major stock assessment region (SAR) by Group from the most recent decade (2016 to 2025), and the previous decade (2006 to 2015). The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Sample sizes are given in Table 16. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

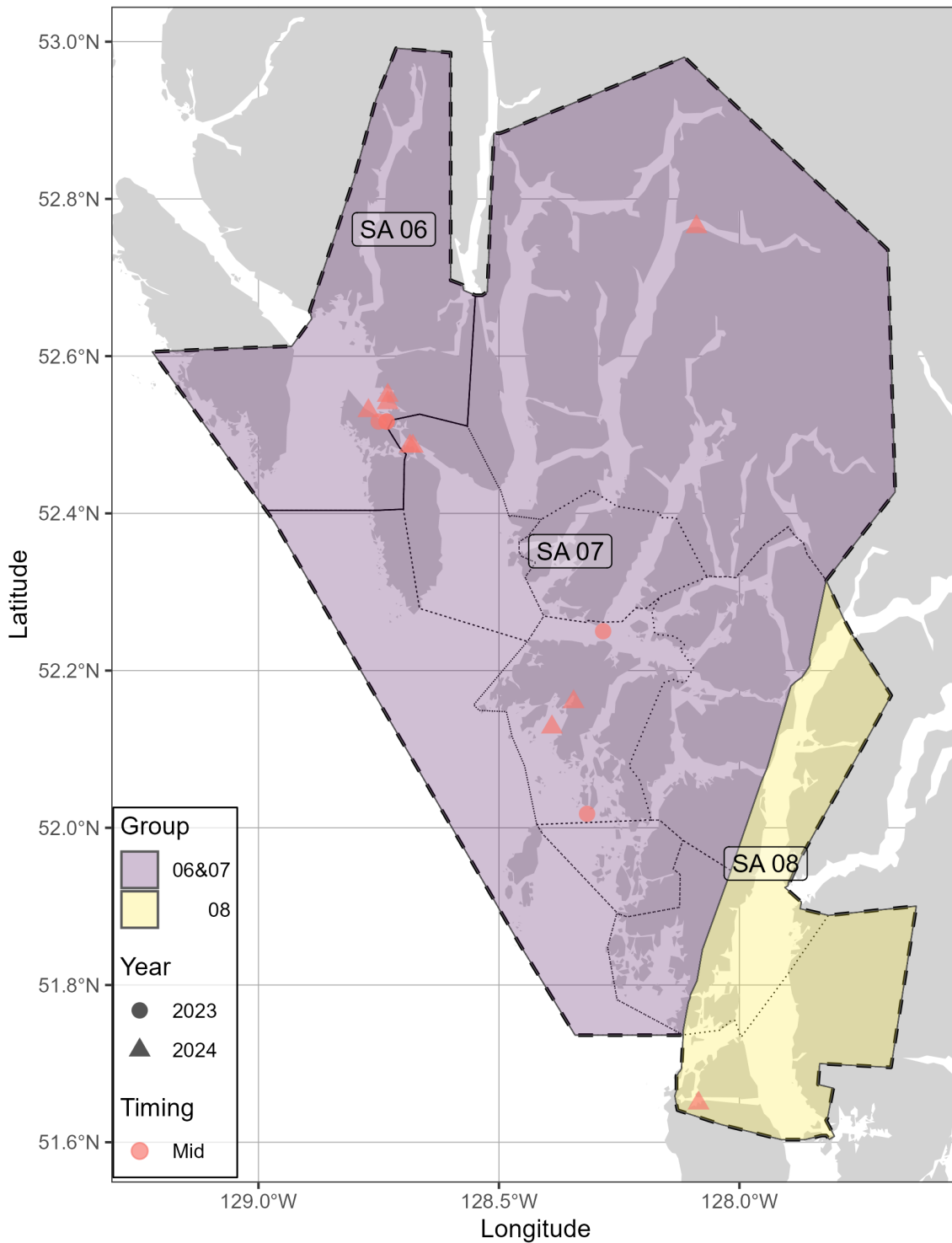


Figure 18. Nearshore genetics sampling in the Central Coast major SAR. Note that timing indicates the collection date: 'Early' is January and February, 'Mid' is March and April, and 'Late' is May, June, and July. In addition, note that genetic samples from 2025 are not yet available. See Figure 2 for legend.

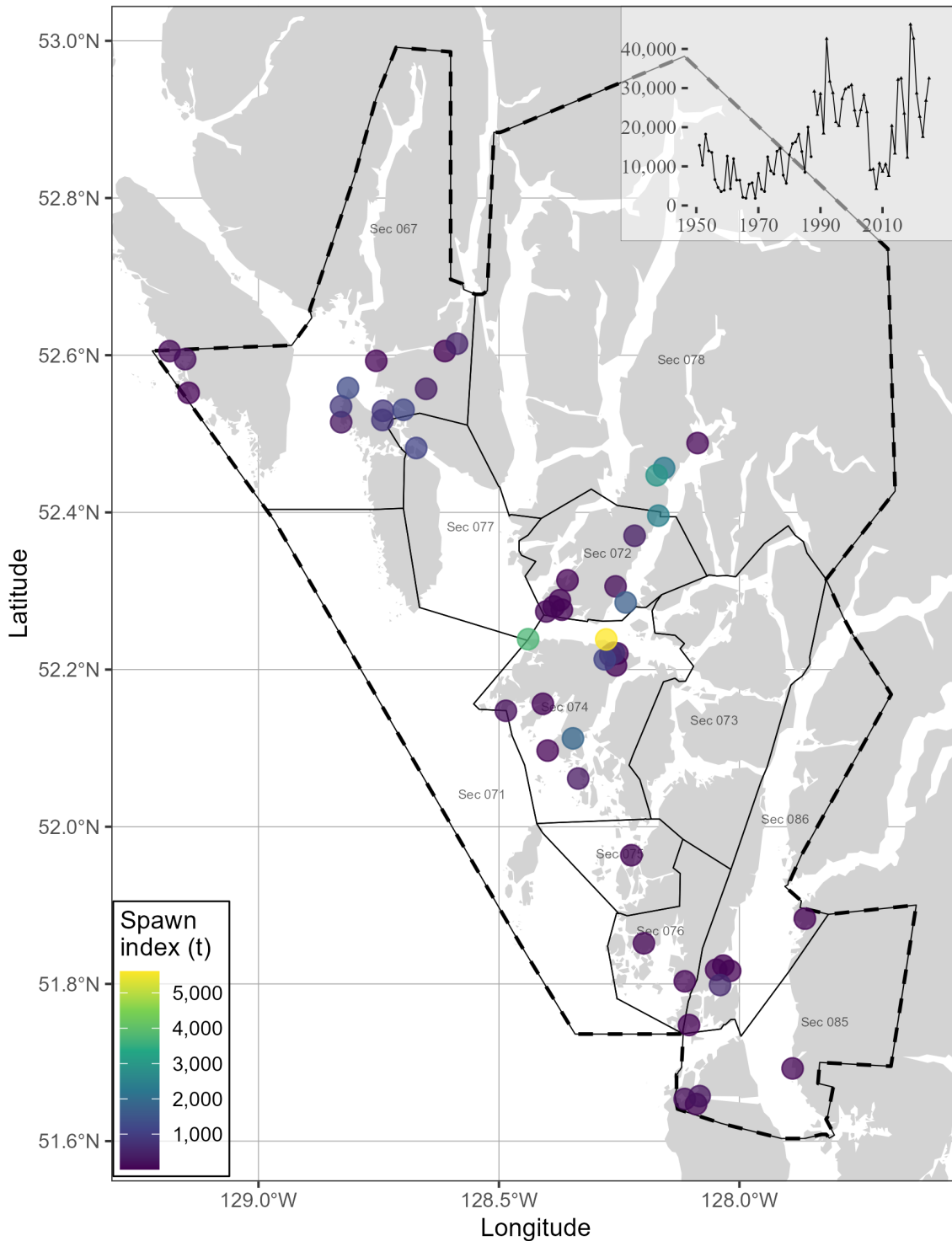


Figure 19. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2025 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

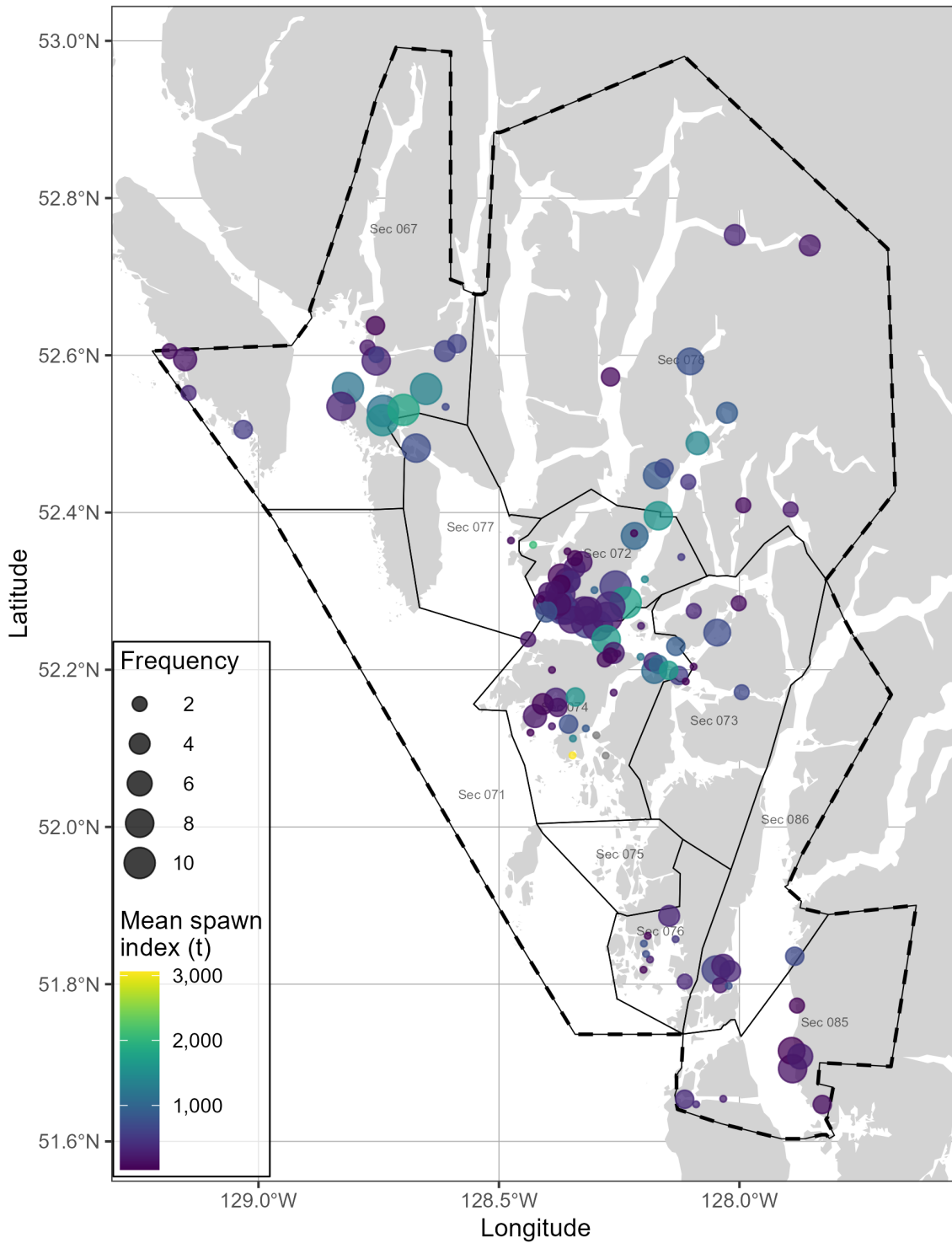


Figure 20. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2015 to 2024 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

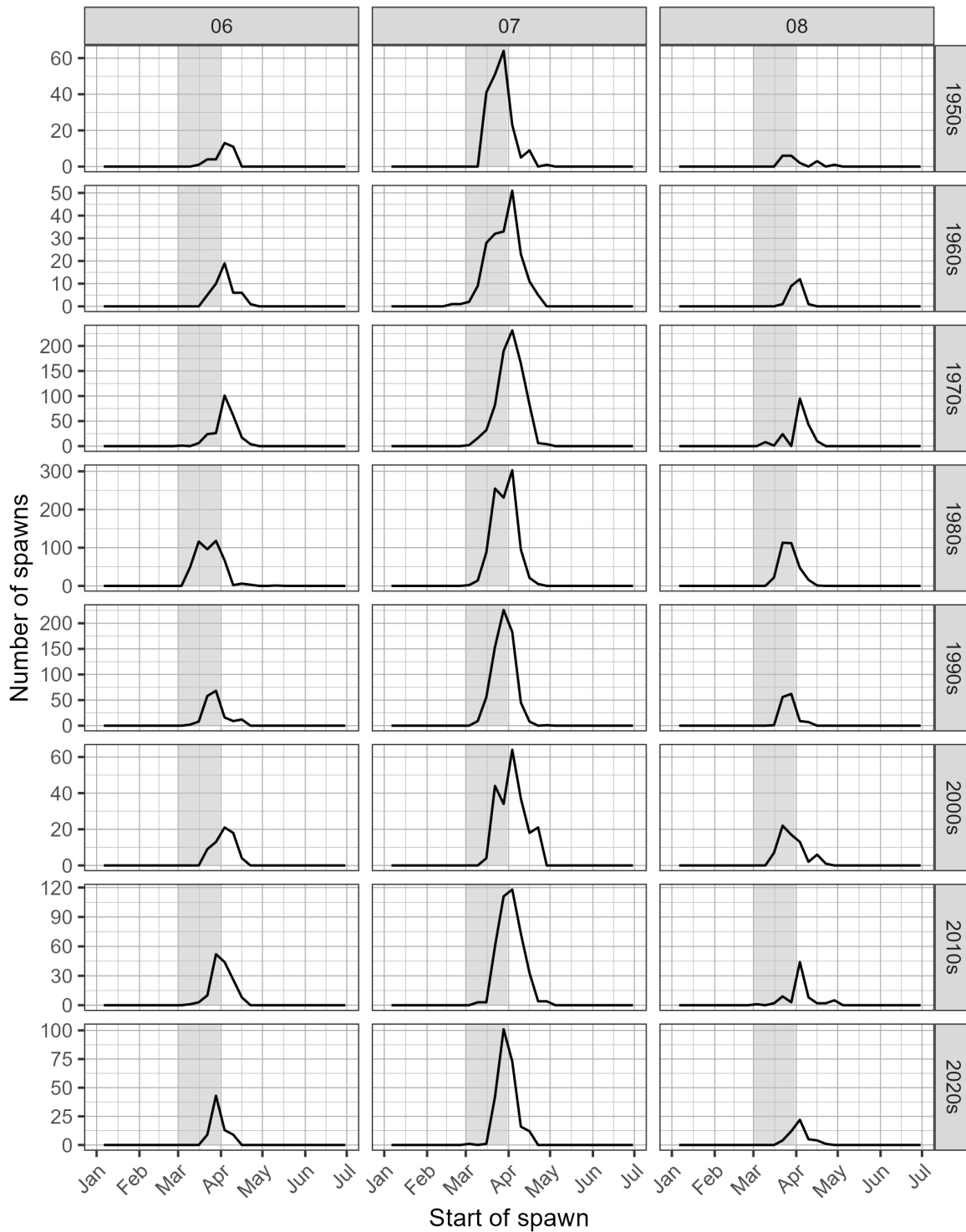


Figure 21. Pacific Herring spawn start date by decade and Statistical Area. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass.

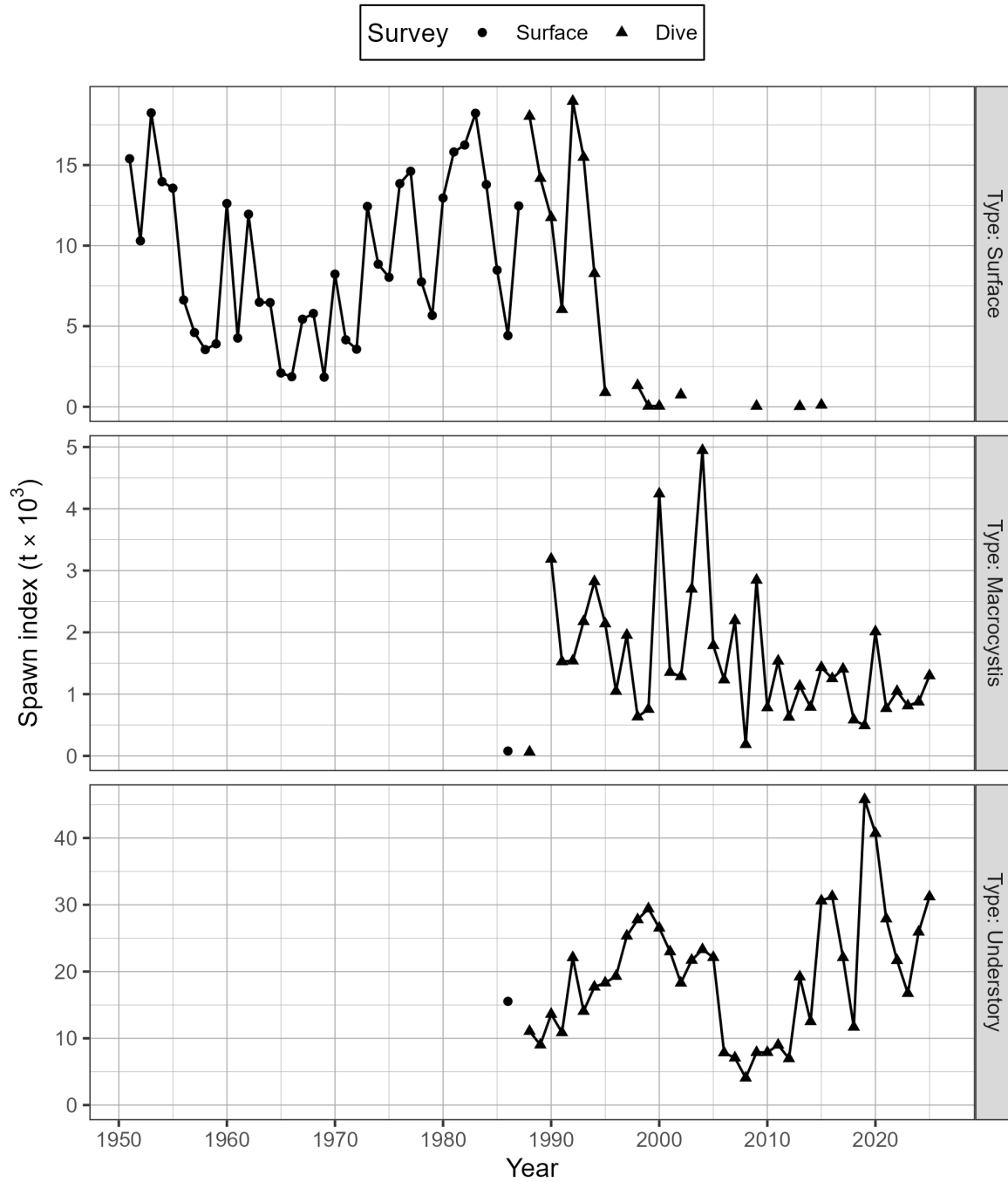


Figure 22. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) by type for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, *Macrocytis* (*Macrocytis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as ‘understory.’ The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

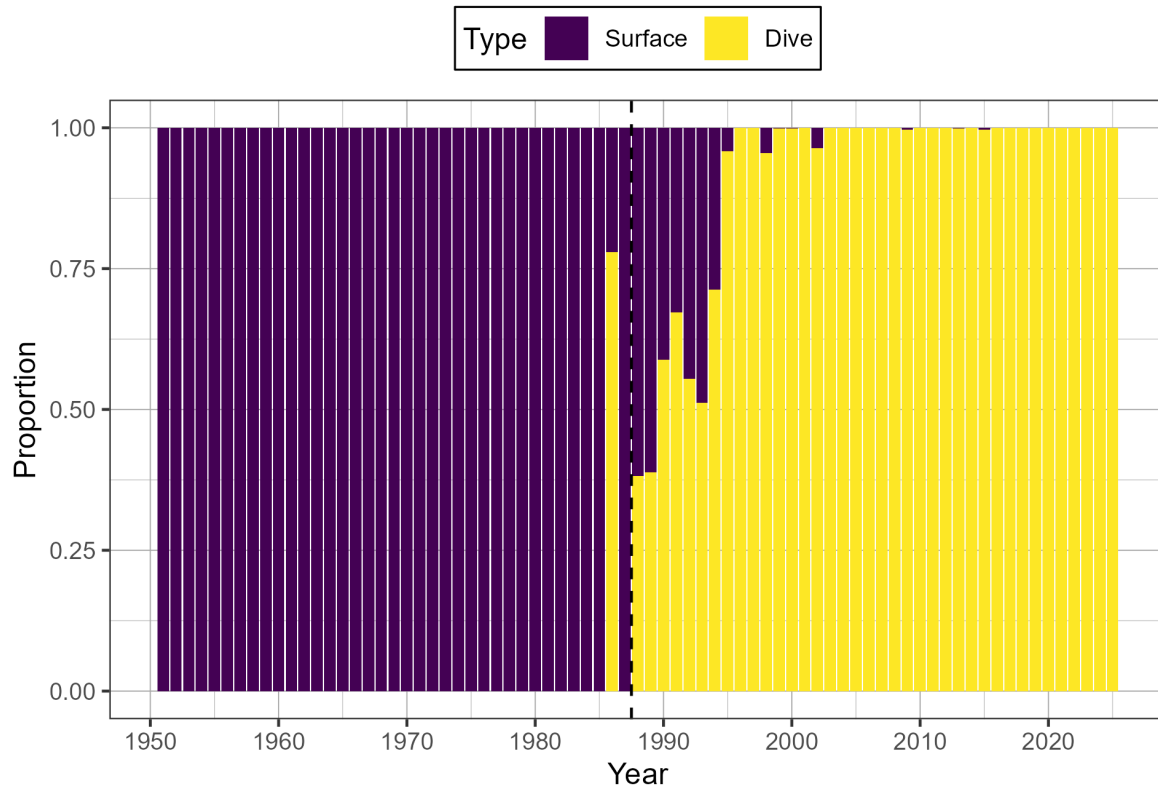


Figure 23. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

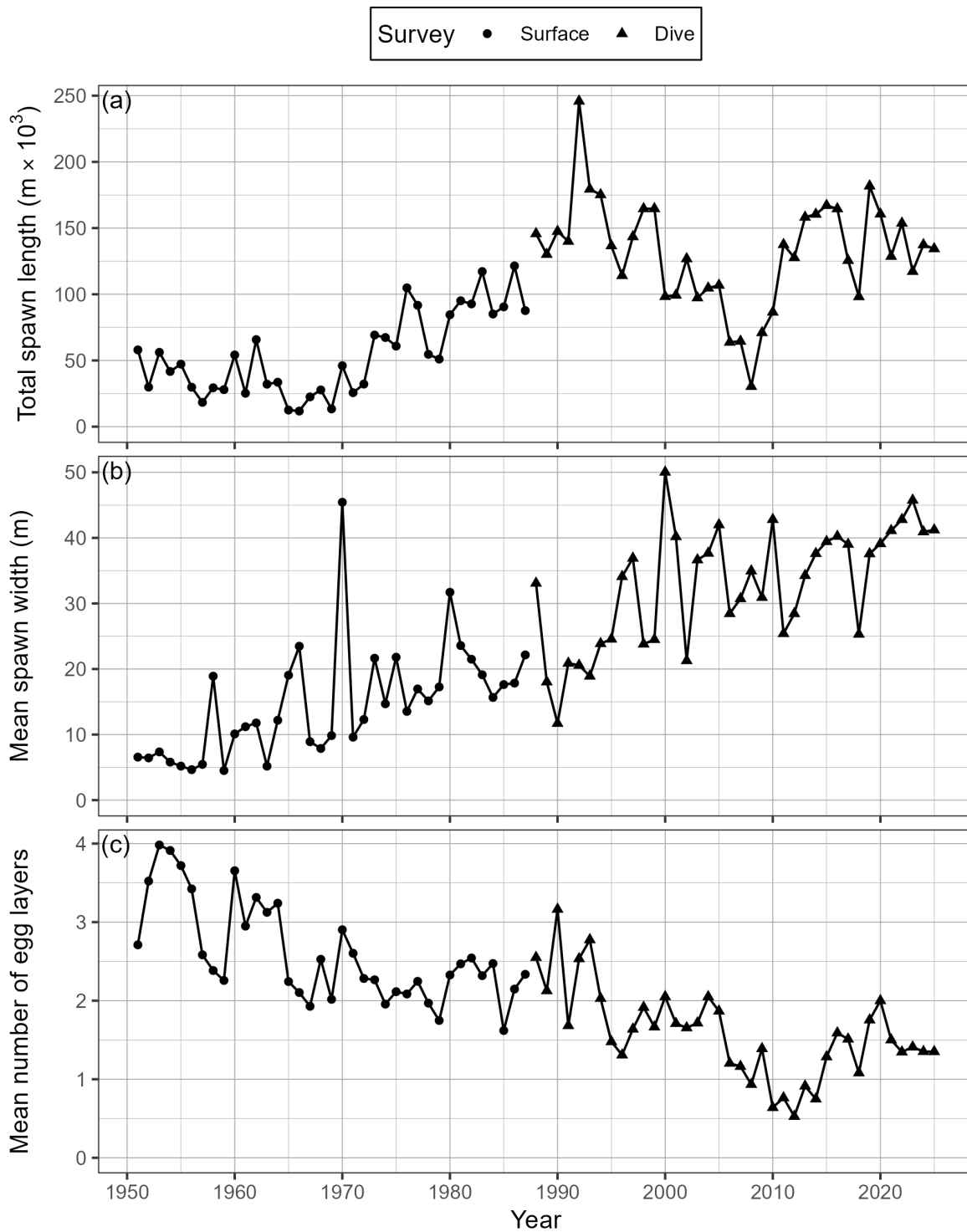


Figure 24. Time series of total spawn length in thousands of metres ($\text{m} \times 10^3$; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

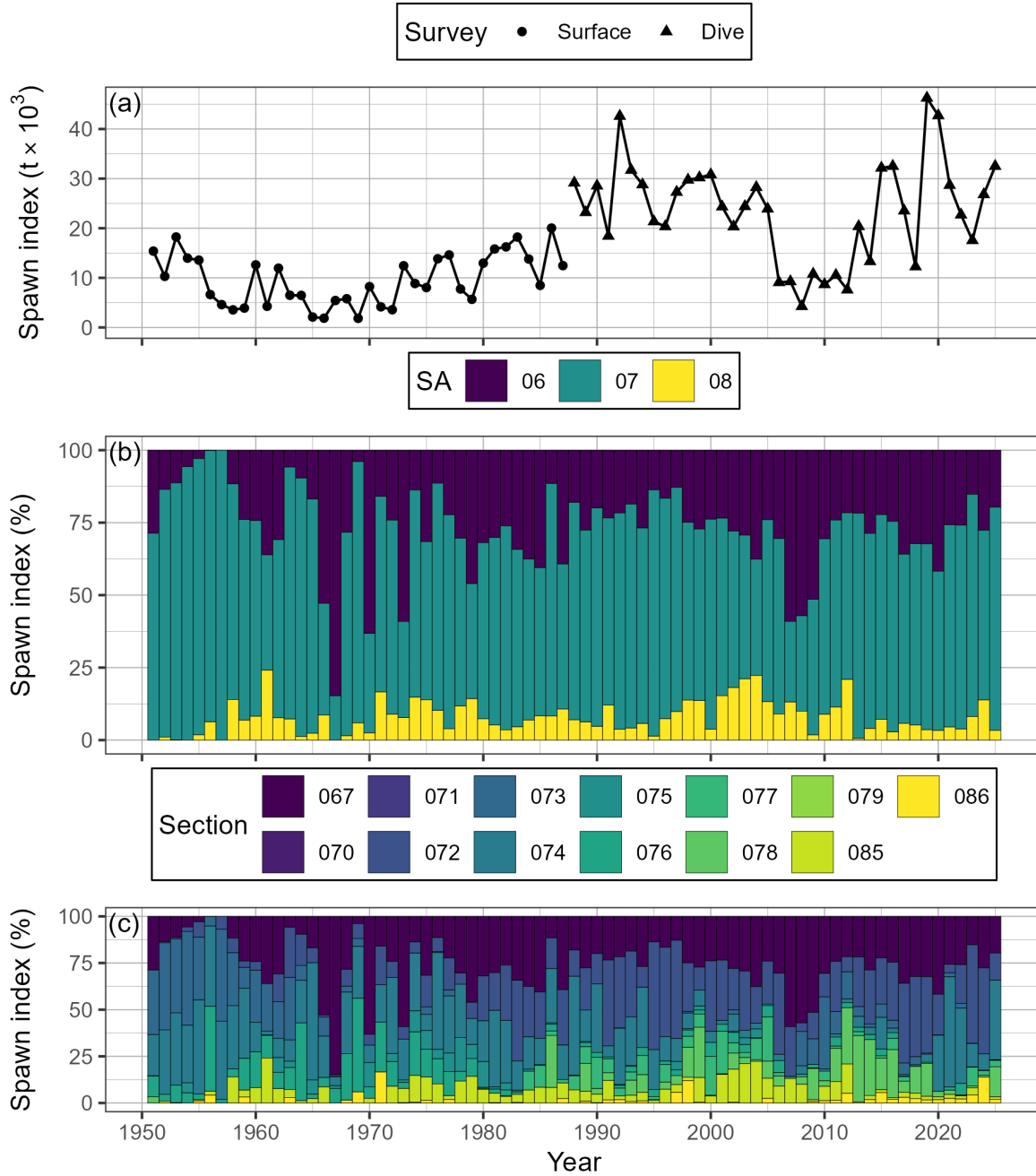


Figure 25. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR; panel a), as well as percent contributed by Statistical Area (SA), and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). Note that spawn surveys in the dive survey period (1988 to 2025) are a combination of surface and dive surveys (Figures 22 and 23). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

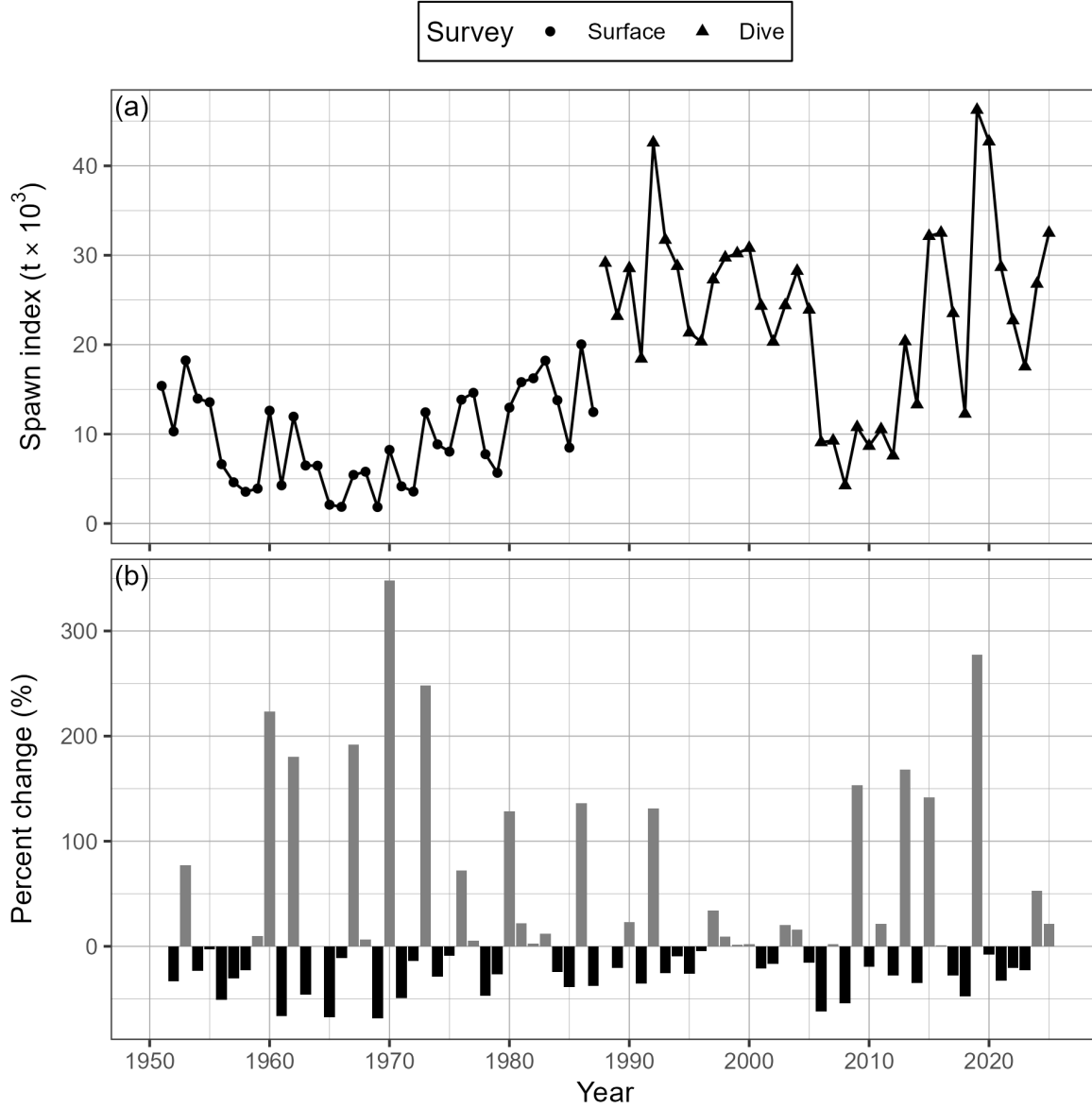


Figure 26. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t . The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Note that spawn surveys in the dive survey period (1988 to 2025) are a combination of surface and dive surveys (Figures 22 and 23).

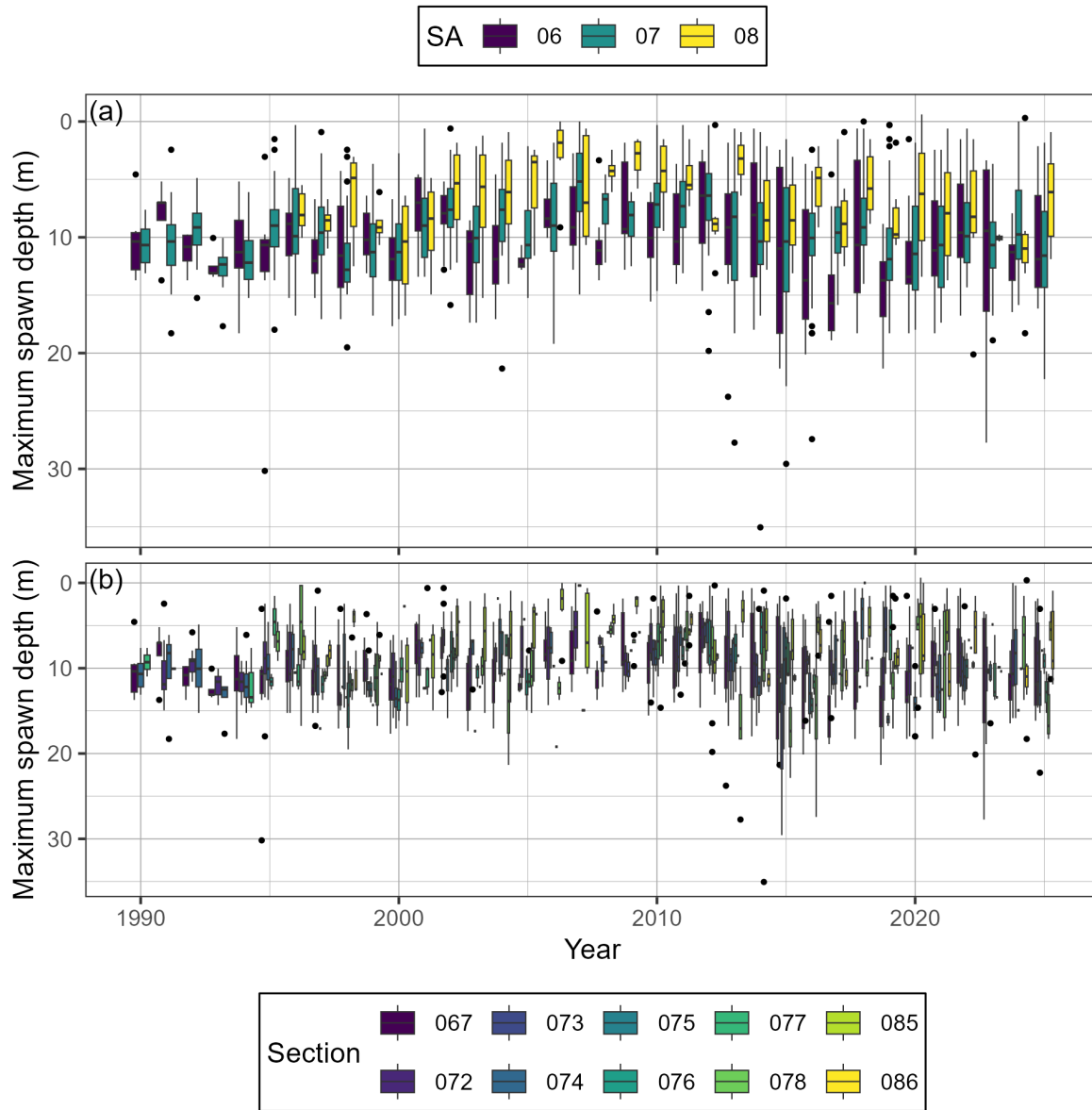


Figure 27. Time series of maximum spawn depth in metres (m) for Pacific Herring from 1990 to 2025 in the Central Coast major stock assessment region (SAR) by Statistical Area (SA; panel a), and Section (b). Note that depth is not corrected to chart datum. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025).

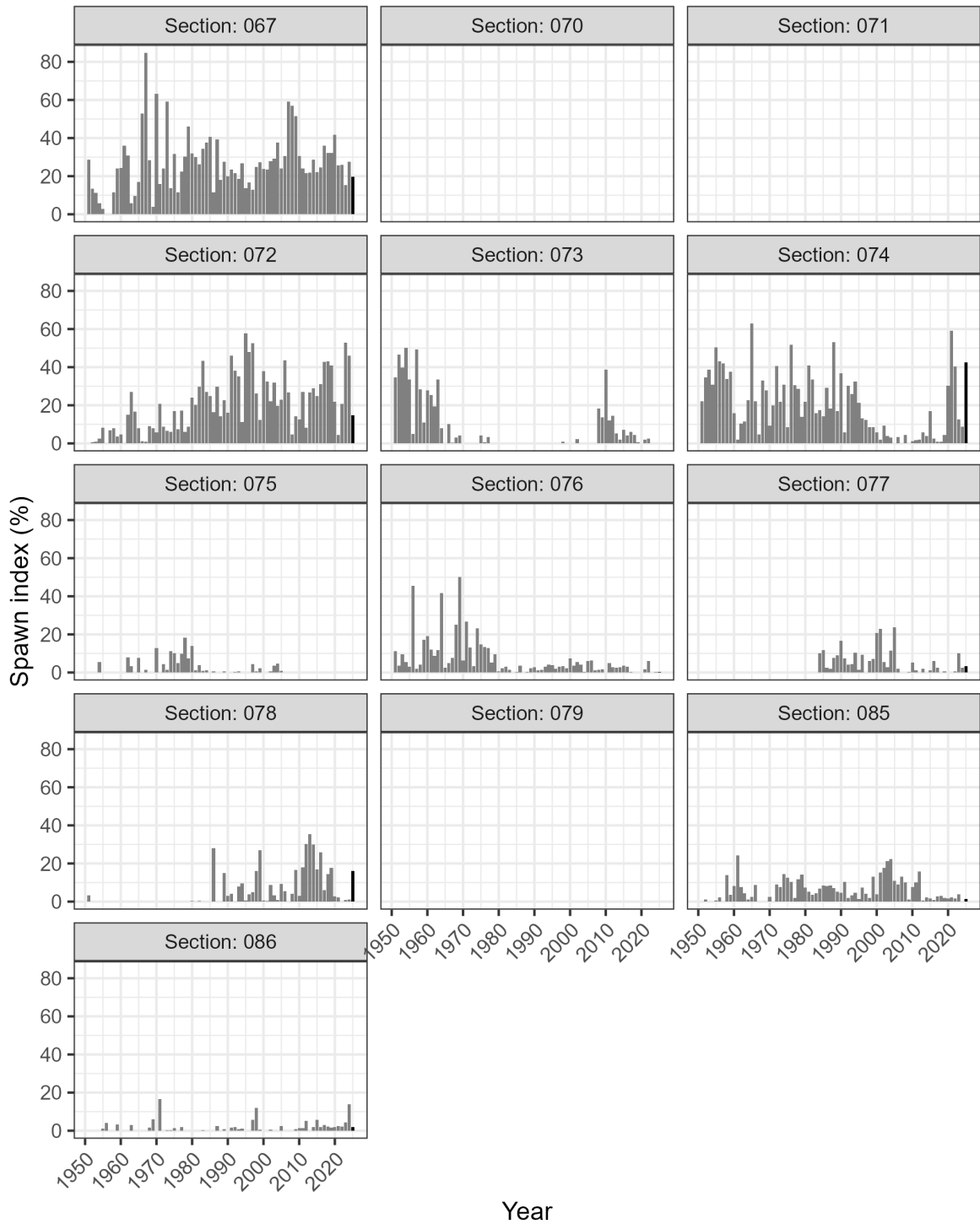


Figure 28. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). The year 2025 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q .

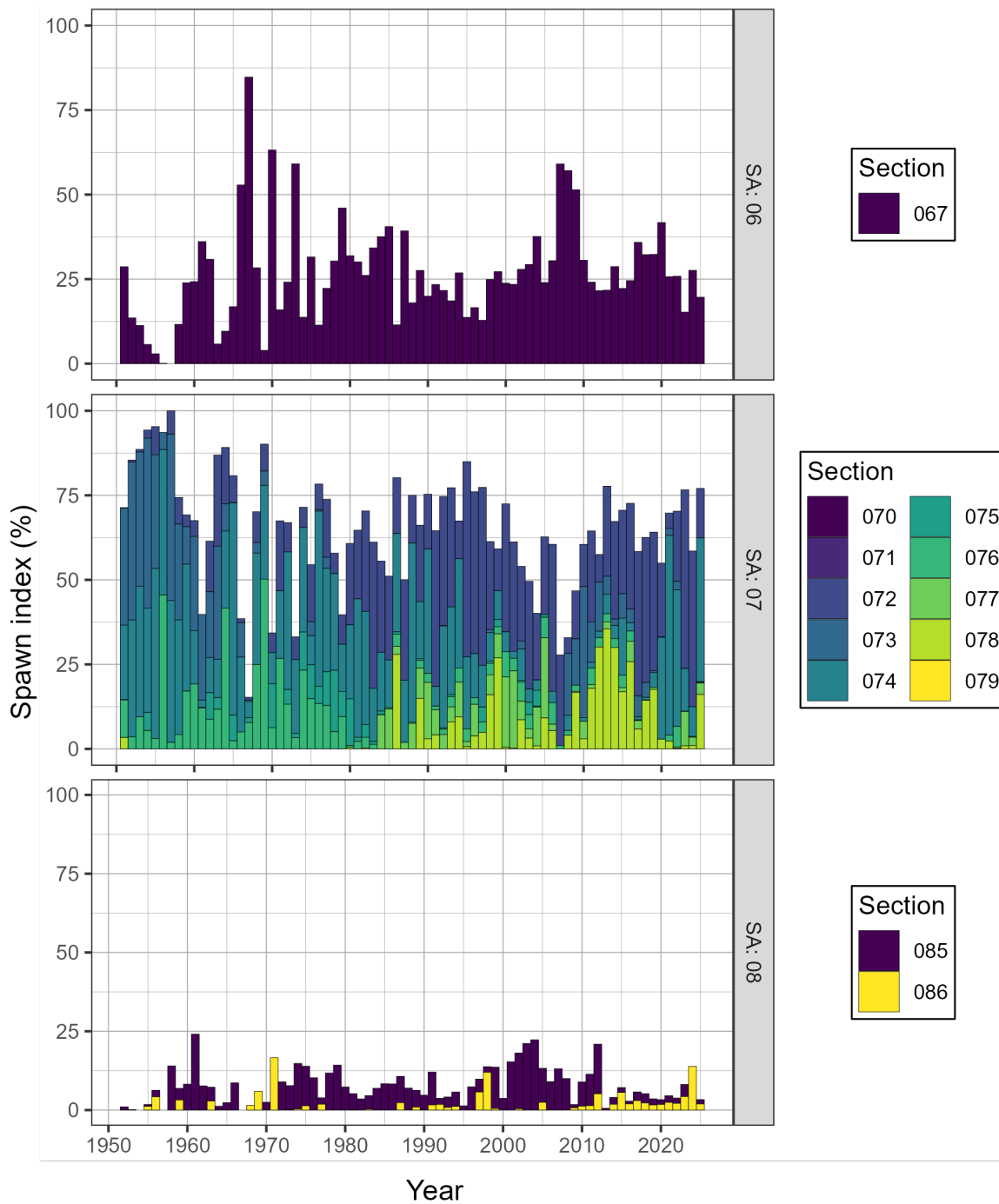


Figure 29. Time series of percent of spawn index by Statistical Area (SA) and Section for Pacific Herring from 1951 to 2025 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

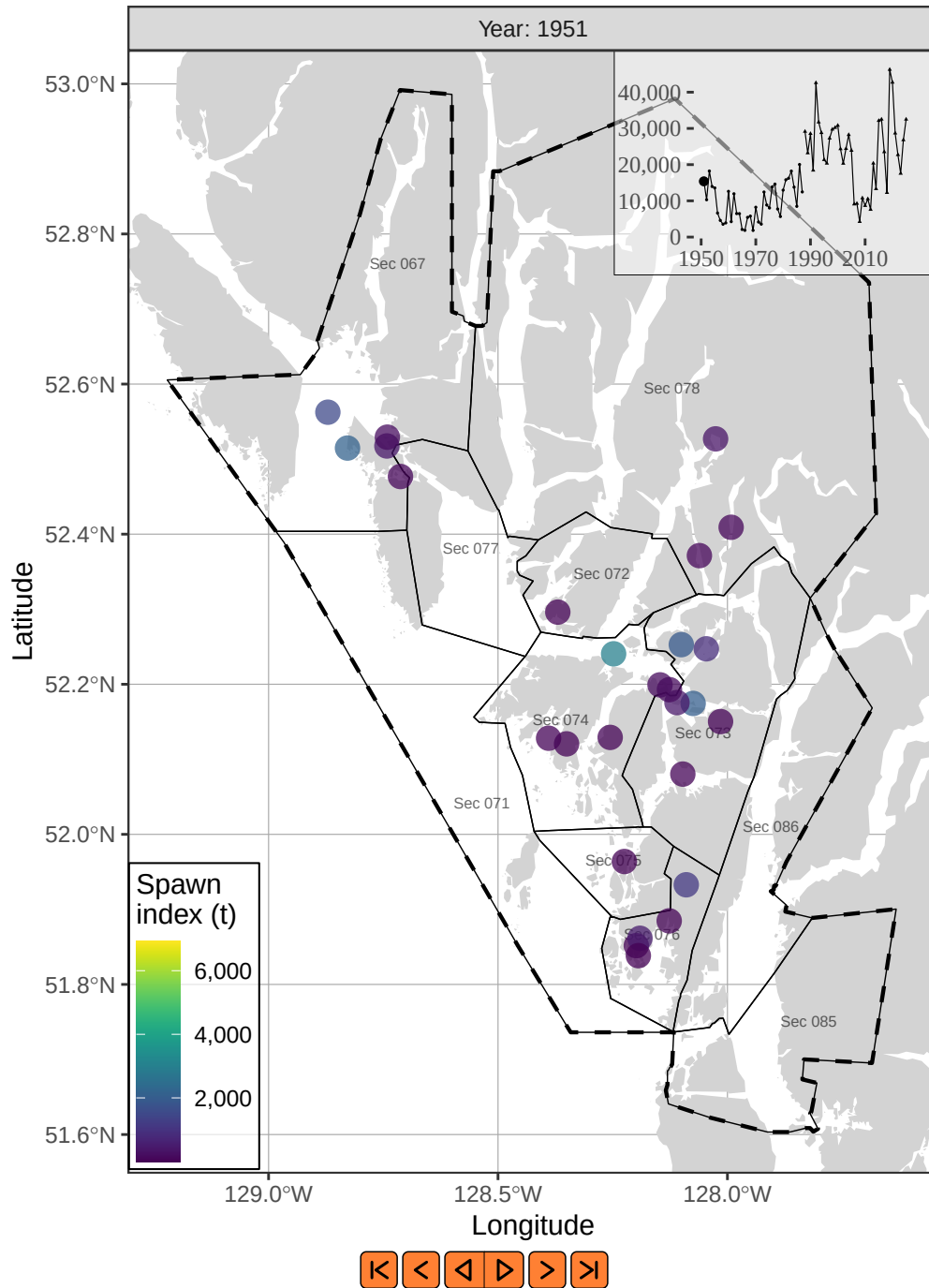


Figure 30. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2025 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2025). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: [download the report](#), open with Adobe, enable Java, and click “play”.